UNIT – I

Fluid Statics, Buoyancy & Floatation

Subject : Fluid Mechanics & Hydraulic Machines (23A03402T)

> Syllabus: Dimensions & units, physical properties of fluids, pressure & its measurement (manometers), Pascal's & hydrostatic laws, buoyancy, metacentre, stability of floating and submerged bodies.

1. Introduction & Dimensions

Fluid = a substance that deforms continuously under the action of a shear stress, however small. (Liquids + gases.)

- Fluid mechanics = fluid statics (fluid at rest) + fluid kinematics (motion without forces) + fluid dynamics (motion with forces).
- Hydraulics deals with practical applications of liquids (mostly water) – pipes, pumps, turbines, channels.

Dimensions & Units

Every physical quantity is expressed in terms of fundamental (primary) dimensions: M, L, T (mass, length, time). All others are derived (secondary).

Units used: SI system – kg, m, s, N, Pa, W. 1 N = 1 kg·m/s^2 , 1 Pa = 1 N/m^2.

Quantity	Dimension	SI unit
Velocity v	$L\,T^{1}$	m/s
Acceleration a	$L\,T^{2}$	m/s^2
Discharge Q	$L^{3}\,T^{1}$	m^3/s
Force F	$M\,L\,T^{2}$	N
Pressure / stress p, τ	$M\,L^{-1}\,T^{2}$	N/m^2 (Pa)
Density ρ	$M\,L^{-3}$	kg/m^3
Specific weight w	$M\,L^{-2}\,T^{2}$	N/m^3
Dynamic viscosity μ	$M\,L^{-1}\,T^{1}$	N·s/m^2 (Pa·s)
Kinematic viscosity ν	$L^{2}\,T^{1}$	m^2/s
Surface tension σ	$M\,T^{2}$	N/m

Quantity	Dimension	SI unit
Work / Energy	$M\,L^2\,T^2$	J
Power P	$M\,L^2\,T^3$	W
Bulk modulus K	$M\,L^{-1}\,T^2$	N/m^2

2. Physical Properties of Fluids

(a) Density, Specific weight, Specific volume, Specific gravity

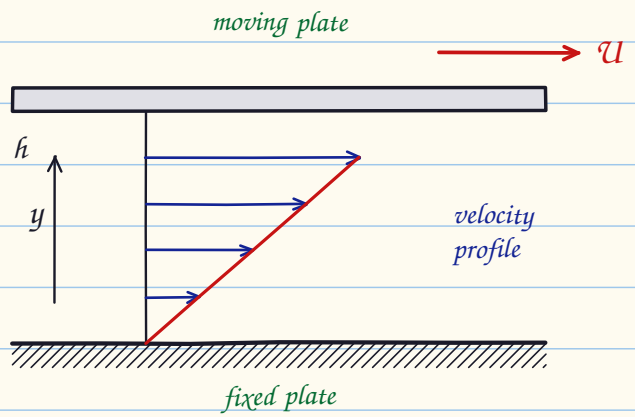
- Mass density ρ = mass / volume (kg/m^3). Water = 1000 kg/m^3 at 4°C ; Mercury = 13600 ; Air ≈ 1.2.
- Specific weight $w = \rho\, g$ = weight/volume. Water = 9810 N/m^3.
- Specific volume $v_s = 1/\rho$ (m^3/kg).
- Specific gravity (relative density) $S = \rho_{fluid} / \rho_{water\ at\ 4°C}$ – dimensionless. S for mercury = 13.6.

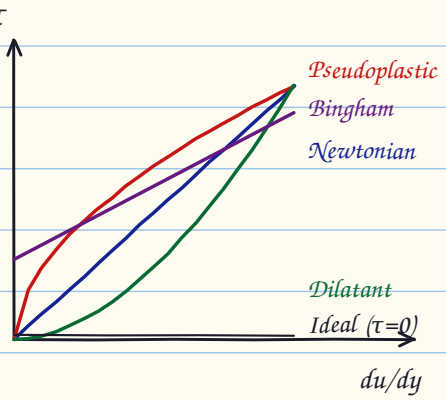
(b) Viscosity

Viscosity is the property of a fluid by virtue of which it offers resistance to relative motion (shear) between its layers. It is due to cohesion and molecular momentum exchange between layers.

Newton's law of viscosity : shear stress is proportional to the rate of shear strain (velocity gradient).

$$\tau = \mu\,(du/dy)$$





- μ = dynamic (absolute) viscosity, $N{\cdot}s/m^2$. 1 poise = 0.1 $N{\cdot}s/m^2$; 1 centipoise = 0.001 $Pa{\cdot}s$ (water ≈ 1 cP at 20°C).
- Kinematic viscosity $\nu = \mu/\rho$ (m^2/s). 1 stoke = 10^{-4} m^2/s.

- *Newtonian fluids* obey the linear law (water, air, oil). *Non-Newtonian* do not : pseudoplastic (paint, blood), dilatant (quicksand, starch), Bingham plastic (toothpaste, sewage sludge).
- *Ideal fluid* : $\mu = 0$ and incompressible. It is only a theoretical concept.

Effect of temperature

- Liquids : μ *decreases* as temperature rises (cohesion falls). Eq : $\mu = \mu_0 / (1 + \alpha t + \beta t^2)$.
- Gases : μ *increases* with temperature (molecular momentum exchange rises). Eq : $\mu = \mu_0 + \alpha t - \beta t^2$.

Significance of viscosity

Viscosity decides the frictional (shear) resistance and head loss in pipes, the lubrication film in bearings, laminar/turbulent nature of flow (Reynolds number), boundary layer and drag on bodies.

> Ex 1. Two plates are 12.5 mm apart. Upper plate moves at 2.5 m/s and needs a shear stress of 5 N/m^2. Find μ. Sol: $\tau = \mu\, U/h \Rightarrow \mu = 5 \times 0.0125 / 2.5 =$ *0.025* *$N{\cdot}s/m^2$* $(= 0.25$ poise$)$.

(c) Compressibility & Bulk modulus

Compressibility $\beta = 1/K$. Bulk modulus $K = -\,dp / (dV/V) = \rho\, dp/d\rho$.

For water $K \approx 2.07 \times 10^9$ N/m^2 (nearly incompressible); for an ideal gas, isothermal $K = p$ and adiabatic $K = k\,p$.

(d) Surface tension

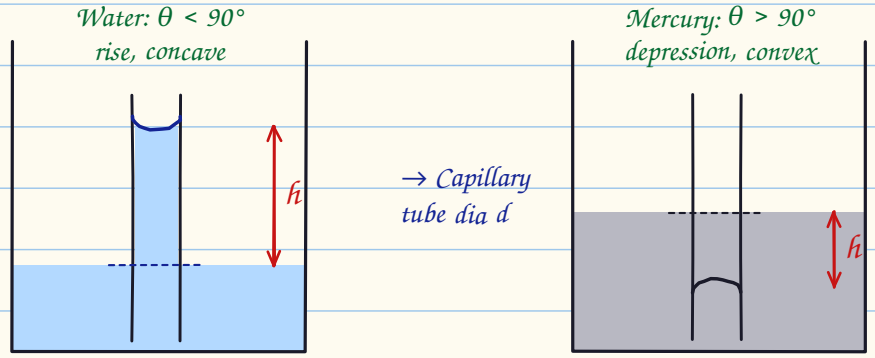
Cohesive force at the free surface/interface makes the liquid surface behave like a stretched elastic membrane. σ = force per unit length (N/m). Water–air at 20°C : $\sigma = 0.0728$ N/m.

For a surface tension force balance (pressure inside curved surface above outside):

Shape	Excess pressure p	Reason
Spherical droplet (dia d)	$p = 4\sigma/d$	one surface
Soap bubble	$p = 8\sigma/d$	two surfaces
Liquid jet (dia d)	$p = 2\sigma/d$	cylinder

(e) Capillarity

Rise or fall of a liquid in a small-bore tube relative to the surrounding level, due to cohesion and adhesion.



$$h = 4\sigma\cos\theta / (\rho g d)$$

- Adhesion > cohesion (water-glass, $\theta \approx 0°$) → rise, concave meniscus. Cohesion > adhesion (mercury-glass, $\theta \approx 130°–140°$) → fall, convex.
- $h \propto 1/d$: the thinner the tube, the higher the rise. Important in soil moisture, plant sap, blotting paper and in the error of manometer/piezometer readings (use tube dia > 12 mm).

Ex 2. Water rise in a 2 mm dia glass tube (σ = 0.0736 N/m, θ = 0°) : h = 4(0.0736)/(1000×9.81×0.002) = 0.294/19.62 = 0.015 m = 15 mm.

(f) Vapour pressure

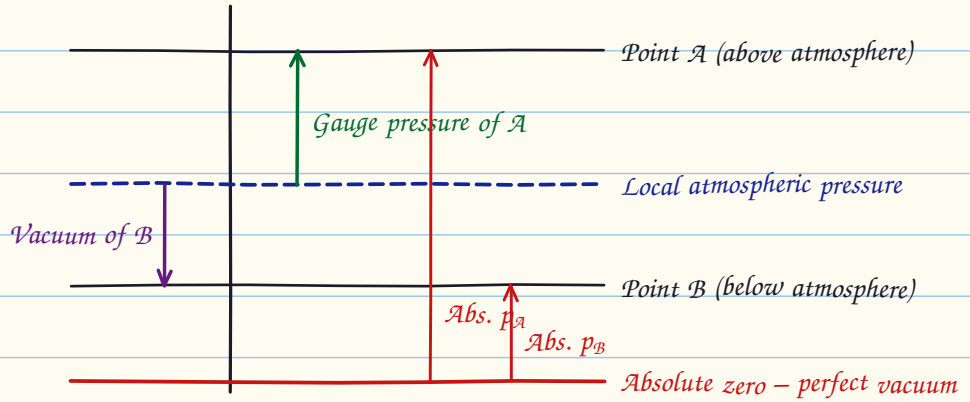
Pressure exerted by the vapour in equilibrium with its liquid at a given temperature. It rises with temperature. Water : 2.34 kPa at 20°C and 101.3 kPa at 100°C.

- A liquid boils when its absolute pressure falls to its vapour pressure.
- If local pressure in a flowing liquid (pump suction, turbine draft tube, venturi throat) falls to p_v, vapour bubbles form and collapse – cavitation : noise, vibration, pitting, loss of efficiency.

3. Pressure & its Measurement

Pressure = normal force per unit area, $p = dF/dA$ (Pa). At a point in a static fluid pressure acts equally in all directions (it is a scalar).

Atmospheric, Gauge, Vacuum and Absolute pressure



$$p_{abs} = p_{atm} + p_{gauge} \quad p_{vac} = p_{atm} - p_{abs}$$

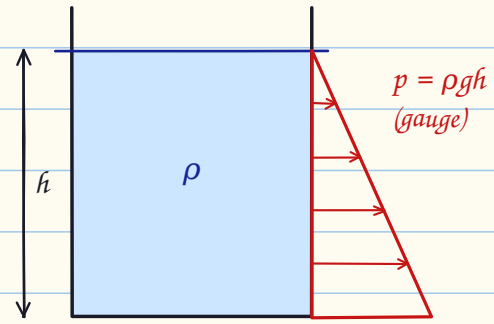
- Standard atmosphere : 1 atm = 101.325 kPa = 760 mm Hg = 10.33 m of water = 1.01325 bar.
- Absolute pressure can never be negative. Gauge pressure is negative at a point below atmosphere (= vacuum or suction pressure).
- Barometer (mercury) measures p_{atm} : $p_{atm} = \rho_{Hg}\, g\, h = 13600 \times 9.81 \times 0.76 = 101.4$ kPa.

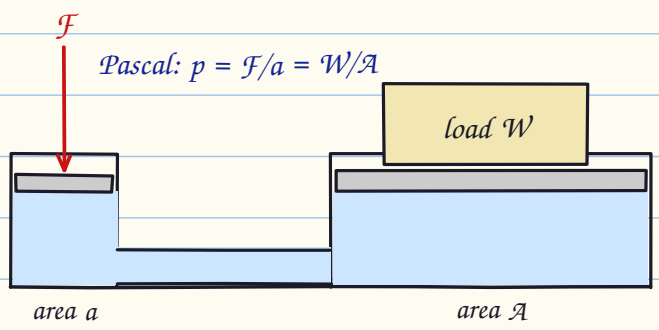
4. Pascal's Law & Hydrostatic Law

Pascal's law : intensity of pressure at a point in a fluid at rest is the same in all directions. A pressure applied to a confined fluid is transmitted undiminished everywhere.

Hydrostatic law : the rate of increase of pressure in the vertically downward direction equals the specific weight of the fluid at that point.

$$dp/dz = \rho\, g \Rightarrow p = \rho\, g\, h \text{ (gauge)}$$





- All points at the same level in a continuous static fluid are at the same pressure (horizontal planes = isobaric).
- Pressure head $h = p/(\rho g)$: the height of fluid column that produces p.
- Hydraulic press / jack / brake : $F/a = W/A \rightarrow$ mechanical advantage $= A/a$.
- Ex. Gauge pressure at 10 m depth in water $= 1000 \times 9.81 \times 10 = 98.1$ kPa ; absolute $= 98.1 + 101.3 =$ 199.4 kPa.

Total pressure & centre of pressure (plane surface)

For a plane surface of area A, centroid at depth $\bar{x}$, inclined at angle θ to the free surface :

$$F = \rho\, g\, A\, \bar{x}$$

$$h_{cp} = \bar{x} + (I_G \sin^2\theta) / (A\, \bar{x})$$

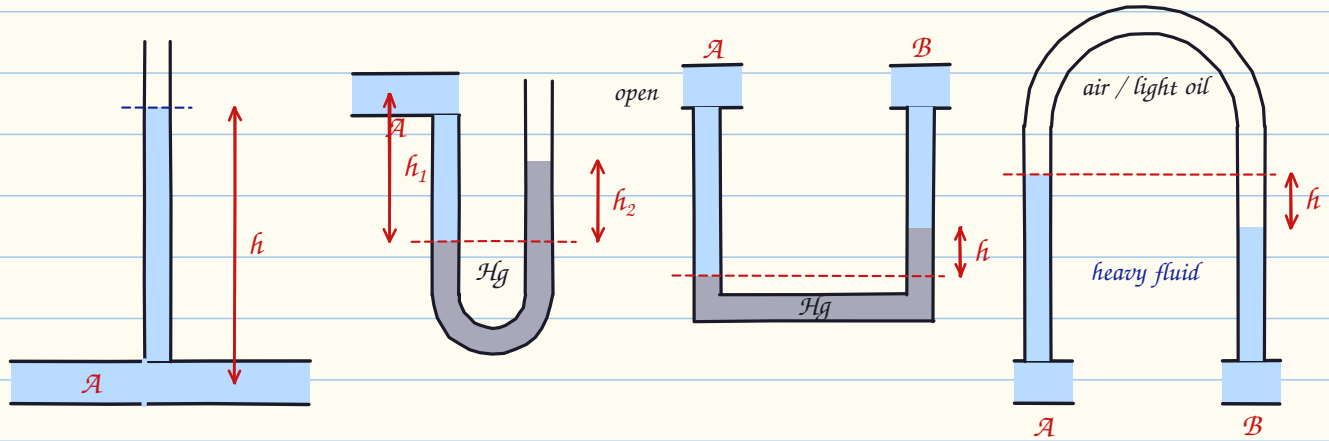
The centre of pressure always lies below the centroid (except for a horizontal surface).

> Ex 4(a). Vertical rectangular gate 2 m wide × 3 m deep, top edge at the free surface of water. $A = 6\ m^2$, $\bar{x} = 1.5$ m. $F = 1000 \times 9.81 \times 6 \times 1.5 =$ 88.3 kN. $h_{cp} = 1.5 + (2 \times 3^3/12)/(6 \times 1.5) = 1.5 + 0.5 =$ 2.0 m from the surface (= 2/3 depth).

5. Measurement of Pressure – Manometers

Manometers measure pressure by balancing it against a column of liquid (gauge liquid). Gauge liquid should be immiscible, of a known density (mercury, water, oil) and a clear meniscus.

Piezometer U-tube Differential Inverted U



(i) Piezometer

Simplest – open vertical tube tapped to the pipe. Only for liquids, only moderate gauge pressure (tube gets too tall), not for gases or negative pressures.

$$p_A = \rho g h$$

(ii) U-tube manometer

The U-tube holds a heavy gauge liquid (Hg, ρ_2) ; pipe liquid has ρ_1. Equate pressures at the common datum level (the interface in the left limb) :

$$p_A + \rho_1 g h_1 = \rho_2 g h_2 \Rightarrow p_A = \rho_2 g h_2 - \rho_1 g h_1$$

- For gas in the pipe, $\rho_1 g h_1$ is negligible $\rightarrow p_A = \rho_2 g h_2$.
- For vacuum (suction) pressure the gauge liquid rises in the pipe-side limb : $p_A = -(\rho_2 g h_2 + \rho_1 g h_1)$.

Ex 3. Pipe carries water. Centre A is 0.10 m above mercury interface (left limb); mercury in open right limb stands 0.25 m higher. $p_A = 13600(9.81)(0.25) - 1000(9.81)(0.1) = 33354 - 981 =$ 32373 $N/m^2 \approx$ 32.4 kPa (gauge).

(iii) Single-column manometer

A U-tube whose one limb is a large reservoir (area A) and the other a narrow tube (area a). Change in reservoir level is tiny, so only one limb is read.

$$p_A = \rho_2 g h_2(1 + a/A) - \rho_1 g (h_1 + a h_2/A) \; [\Delta h = a h_2/A]$$

(iv) Differential manometers (pressure DIFFERENCE between two points)

U-tube differential : A and B contain fluid ρ_1, mercury ρ_g deflects by h :

$$p_A - p_B = g h (\rho_g - \rho_1)$$

If pipes are at different levels, add $\rho_1 g (z_B - z_A)$ – i.e., measure all heads from one datum.

Inverted U-tube : used for light liquids/low differences; light fluid (air or oil, ρ_3) at top :

$$p_A - p_B = \rho_1 g h_1 - \rho_2 g h_2 - \rho_3 g h_3 \; (\text{air on top} : p_A - p_B = \rho g h)$$

Ex 3(a). Differential U-tube, pipes A and B at the same level carry water; mercury deflection $h = 0.2$ m. $p_A - p_B = g h(\rho_{Hg} - \rho_w) = 9.81(0.2)(13600 - 1000) = 24721$ Pa = 24.7 kPa (A is at the higher pressure).

(v) Micro-manometers & mechanical gauges

Inclined-tube and micromanometers amplify the reading for small pressures. Mechanical gauges (Bourdon tube, diaphragm, dead-weight tester) and pressure transducers for high pressure or remote reading.

> Method to solve any manometer problem : start from the point of known pressure, go step by step adding $\rho g h$ when moving DOWN and subtracting $\rho g h$ when moving UP, finish at the unknown point.

6. Buoyancy and Floatation

When a body is immersed (wholly or partly) in a fluid, it experiences an upward force – *buoyant force* F_B – equal to the weight of the fluid displaced (Archimedes' principle).

$$F_B = \rho_{fluid}\, g\, V_{displaced}$$

- Buoyant force acts vertically upward through the *centre of buoyancy* B = centroid of the displaced volume.
- Body weight W acts downward through the *centre of gravity* G.
- Conditions : $W > F_B$ sinks ; $W = F_B$ floats in equilibrium (partially or fully submerged) ; $W < F_B$ rises to the surface.
- For a floating body, fraction submerged = ρ_{body}/ρ_{fluid}. Iceberg (917 kg/m^3) in sea water (1025) : ≈ 89.5 % below water.
- Applications : ships, submarines, balloons/airships, hydrometers (density measurement), life jackets, buoys, pontoons.

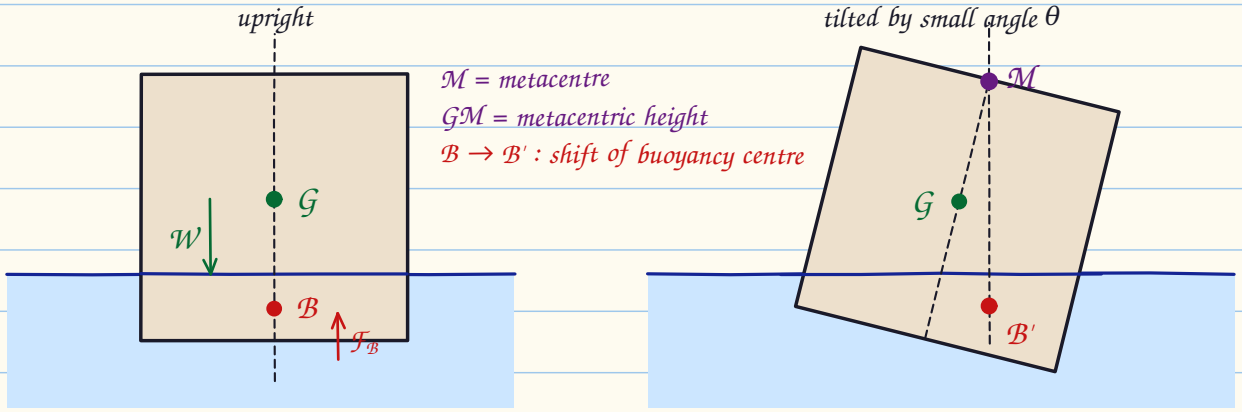
Equilibrium of submerged bodies

Position	Condition	Equilibrium
B above G	restoring couple on tilt	Stable
B below G	overturning couple	Unstable
B coincides with G	no couple (homogeneous body)	Neutral

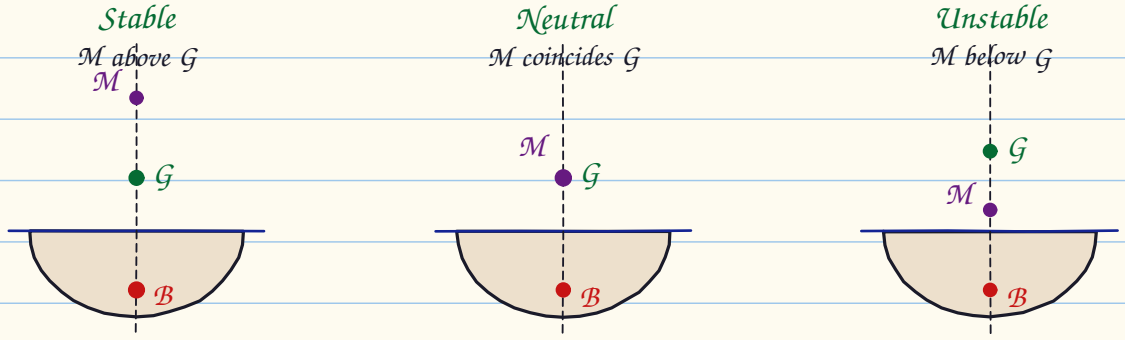
A completely submerged body keeps the same B and G whatever the tilt – stability depends only on their relative position (e.g. submarine, balloon).

7. Metacentre & Stability of Floating Bodies

In a floating body the shape of displaced volume changes when it tilts, so B shifts to B′. The vertical line through B′ cuts the original axis (BG line) at M, the metacentre.



- Metacentre M : the point about which the body starts oscillating when tilted by a small angle; point of intersection of the line of action of F_B (after tilt) with the original vertical axis.
- Metacentric height GM = distance between G and M.



Stable : M above G (GM positive) Neutral : M = G Unstable : M below G

Derivation of BM (metacentric radius)

Tilt the body by small angle θ. A wedge on one side goes into the water, an equal wedge emerges on the other side, so the buoyancy force shifts through BB′.

- Moment of the wedge forces about the axis = $F_B \times BB'$ ... a strip at distance x of width dx has volume $x\,\theta\,dA$; the moment of the wedge pair = $\rho\, g\, \theta \int x^2\, dA = \rho\, g\, \theta\, I$.
- Also moment = $F_B \times BB' = \rho\, g\, V \times BM\,\theta$ (since $BB' = BM \tan\theta \approx BM\,\theta$).

$$BM = I / V \quad GM = BM - BG = I/V - BG$$

- I = second moment of the waterline plane area about the longitudinal axis through its centroid (the axis of tilt) ; V = volume of liquid displaced.
- For rectangular waterplane ($L \times b$) : $I = L\,b^3/12$; for circular (dia d) : $I = \pi d^4/64$.
- If G is below B (BG measured downward) use $GM = BM + BG$.

Ex 4. Rectangular pontoon 10 m × 7 m × 3 m deep, draught 1.5 m (homogeneous, so G at 1.5 m above base). $I = 10\times7^3/12 = 285.8\ m^4$; $V = 10\times7\times1.5 = 105\ m^3$; $BM = 2.72$ m ; $B = 0.75$ m above base ⇒ $BG = 0.75$ m ; $GM = 2.72 - 0.75 = 1.97$ m (+ve) → STABLE.

Ex 5. Wooden cylinder, $d = 0.6$ m, height 1.0 m, $S = 0.6$, floats with axis vertical in water. Draught = $0.6 \times 1.0 = 0.6$ m ⇒ $B = 0.3$ m from base ; $G = 0.5$ m ⇒ $BG = 0.2$ m. $I = \pi(0.6)^4/64 = 0.00636\ m^4$; $V = (\pi/4)(0.6)^2(0.6) = 0.1696\ m^3$; $BM = 0.0375$ m. $GM = 0.0375 - 0.2 = -0.1625$ m → UNSTABLE (will not float upright; it floats with the axis horizontal).

Experimental determination of GM

A known weight w is moved across the deck through a distance x, causing a heel θ. Taking moments of W about G :

$$GM = w\,x / (W \tan\theta)$$

Rolling period

For small oscillations the time period of roll is

$$T = 2\pi \sqrt{(k^2 / (g \cdot GM))}\ (k = \text{radius of gyration})$$

Larger GM → shorter period : the ship is stiff (sharp, uncomfortable return). Small GM → slow roll, tender ship, closer to capsizing.

Applications & stability analysis

- Ships are designed with GM about 0.3 – 1.5 m ; low G by placing ballast/heavy cargo in the bottom, wide beam to increase I.
- Righting moment = $W \times GM \sin\theta$ (valid only for small $\theta \approx < 15°$). At large angles the GZ (righting arm) curve is used.
- Submarines : on the surface stability as a floating body (M above G); submerged – needs B above G, so heavy equipment is kept low.
- Hydrometers, buoys and floating docks are all designed using the GM criterion.

8. Quick Formula Sheet – Unit I

Quantity	Formula
Specific weight, S.G.	$w = \rho g$; $S = \rho/\rho_w$
Newton's law	$\tau = \mu\ du/dy$; $\nu = \mu/\rho$
Excess pressure	droplet $4\sigma/d$, bubble $8\sigma/d$, jet $2\sigma/d$
Capillary rise	$h = 4\sigma\ \cos\theta/(\rho\ g\ d)$
Pressure	$p = \rho\ g\ h$; $p_{abs} = p_{atm} + p_{gauge}$
Manometer	$p_A = \rho_2 g h_2 - \rho_1 g h_1$
Differential	$p_A - p_B = g\ h\ (\rho_g - \rho_1)$
Buoyancy	$F_B = \rho\ g\ V_{disp}$
Metacentre	$BM = I/V$; $GM = BM - BG$; $T = 2\pi\sqrt{(K^2/g\ GM)}$

9. Short Answers – Quick Revision

- *Is mercury a better manometer liquid than water?* Yes – high density gives a short column, low vapour pressure, clear meniscus, does not wet glass.
- *Why does a ship stay upright?* A tilt shifts B outward so F_B and W form a restoring couple whenever M lies above G.
- *Why use a wide hull and low ballast?* Wide hull ↑ I (and BM) ; low ballast lowers G ; both ↑ GM.
- *Difference between gauge and absolute pressure?* Gauge is measured relative to local atmosphere, absolute relative to perfect vacuum.
- *Why is Pascal's law useful?* Basis of hydraulic jacks, presses, brakes, lifts, excavators (force multiplication by area ratio).
- *Why is centre of pressure below centroid?* Pressure increases with depth, so the resultant acts closer to the deeper part.
- *What is a meniscus error?* Capillary rise/depression in a narrow gauge tube – avoided by using tube dia ≥ 12 mm or applying correction.
- *Vapour pressure and cavitation :* if $p \le p_v$ the liquid flashes to vapour; bubbles collapse on reaching high-pressure zones.

10. Important Questions

- Define viscosity. State Newton's law and explain the effect of temperature on viscosity of liquids and gases.
- Derive the expression for capillary rise/fall in a tube. Why does mercury show depression?
- Distinguish between absolute, gauge and vacuum pressure. Derive the hydrostatic law.
- Describe the differential U-tube and inverted U-tube manometers with neat sketches; derive the expressions.
- Define buoyancy, centre of buoyancy and metacentre. Derive $BM = I/V$.
- Explain the experimental determination of metacentric height. State the conditions of stability of floating & submerged bodies.
- Numerical : manometer problems, GM of rectangular/cylindrical vessels, force on submerged plane surfaces.

UNIT – II

Fluid Kinematics, Dynamics & Pipe Flow

Subject : Fluid Mechanics & Hydraulic Machines (23A03402T)

> Syllabus: Types of flow, continuity equation, circulation & vorticity, streamline/pathline/streakline, stream & potential functions, flow net, source-sink-doublet-vortex, Euler's & Bernoulli's equations, momentum equation, pipe bend, Reynolds experiment, Darcy-Weisbach, minor losses, series/parallel pipes, TEL & HGL.

1. Fluid Kinematics – Introduction

Kinematics = study of fluid motion (velocity, acceleration, flow pattern) *without* considering the forces that cause it.

Types of flow

Classification	Type 1	Type 2
Time	Steady : $\partial()/\partial t = 0$ at a point	Unsteady : properties change with t
Space	Uniform : velocity same at all points ($\partial V/\partial s = 0$)	Non-uniform : V varies along flow
Nature	Laminar : orderly layers, Re small	Turbulent : eddies, mixing, Re large
Density	Incompressible : ρ = const (liquids)	Compressible : ρ varies (gases at high Mach no.)
Rotation	Irrotational : fluid particles do not spin	Rotational : particles rotate about own axes
Dimension	1-D : $V = f(s)$ (pipe average)	2-D : $u(x,y)$, $v(x,y)$; 3-D general

- Ex : flow through a pipe at constant Q = steady uniform ; flow through a tapering pipe at constant Q = steady non-uniform ; pipe with varying Q & constant diameter = unsteady uniform.

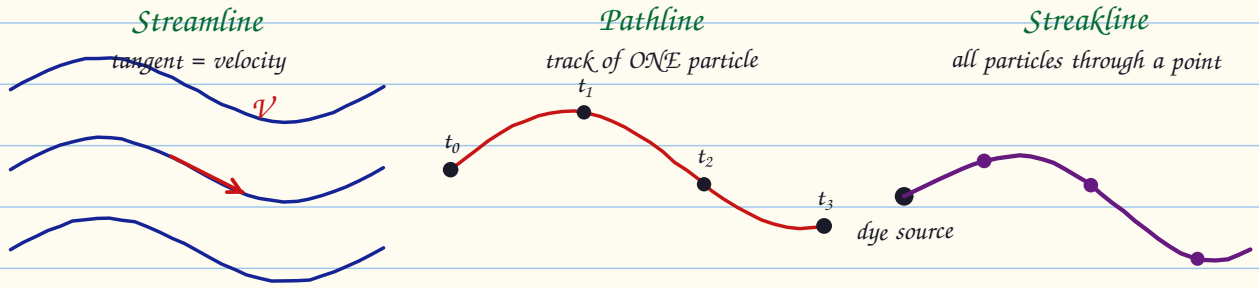
Acceleration of a fluid particle

Velocity $V = f(x,y,z,t)$. Acceleration in x-direction :

$$a_x = \partial u/\partial t + u\,\partial u/\partial x + v\,\partial u/\partial y + w\,\partial u/\partial z$$

- $\partial u/\partial t$ = local (temporal) acceleration – zero for steady flow. Remaining terms = convective acceleration – zero for uniform flow.

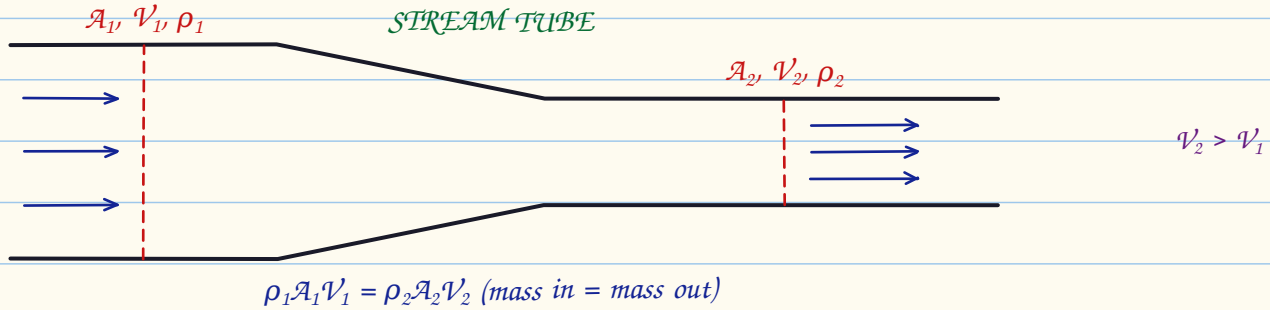
Streamline, Pathline, Streakline, Stream tube



- **Streamline** : imaginary curve whose tangent at every point gives the direction of velocity there. In steady flow streamlines never cross and no flow crosses them. Equation : $dx/u = dy/v = dz/w$.
- **Pathline** : actual track followed by one fluid particle over a time interval. **Streakline** : locus of all particles that have passed through a fixed point (dye/smoke filament).
- For steady flow all three coincide. **Stream tube** : tube formed by a bundle of streamlines through a small closed curve – behaves like a solid pipe wall.

2. Continuity Equation

Conservation of mass : mass entering a stream tube per second = mass leaving (steady flow).



$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$ (mass in = mass out)

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2 \text{ incompressible : } Q = A_1 V_1 = A_2 V_2$$

- For a branching pipe : $Q_1 = Q_2 + Q_3$.
- 3-D differential form (compressible, unsteady) : $\partial\rho/\partial t + \partial(\rho u)/\partial x + \partial(\rho v)/\partial y + \partial(\rho w)/\partial z = 0$

$$\text{Incompressible : } \partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0 \text{ (2-D : } \partial u/\partial x + \partial v/\partial y = 0)$$

Ex 1. A 300 mm pipe reduces to 150 mm. $V_1 = 2$ m/s. $Q = (\pi/4)(0.3)^2(2) = 0.1414$ m^3/s ; $V_2 = Q/A_2 = 0.1414/0.017671 = 8$ m/s (velocity $\propto 1/D^2$).

Ex 2. Is $u = 2x$, $v = -2y$ a possible incompressible flow? $\partial u/\partial x + \partial v/\partial y = 2 - 2 = 0$ ⇒ Yes, continuity is satisfied.

3. Rotation, Vorticity & Circulation

For 2-D flow, angular velocity of a fluid element about z-axis :

$$\omega_z = \tfrac{1}{2}\,(\partial v/\partial x - \partial u/\partial y)$$

- Vorticity $\zeta = 2\omega_z = \partial v/\partial x - \partial u/\partial y$ (twice the rotation).
- Circulation $\Gamma = \oint V \cdot ds$ (line integral of the tangential velocity round a closed curve). By Stokes' theorem $\Gamma = \zeta \times A$ (A = enclosed area).
- **Condition for irrotational flow** : $\omega_z = 0 \Rightarrow \partial v/\partial x = \partial u/\partial y$ (3-D : curl $V = 0$). Then $\Gamma = 0$ around any closed curve within the fluid.

Stream function ψ (2-D, incompressible)

Defined so that continuity is automatically satisfied :

$$u = \partial\psi/\partial y \quad v = -\partial\psi/\partial x$$

- ψ = constant along a streamline ; flow between two streamlines $Q = \psi_2 - \psi_1$ (per unit depth).
- Exists for any 2-D incompressible flow (rotational or not). For irrotational flow $\nabla^2\psi = 0$ (Laplace).

Velocity potential function φ

Exists only for irrotational flow :

$$u = -\partial\varphi/\partial x \; v = -\partial\varphi/\partial y$$

- φ = constant gives an equipotential line. Continuity $\Rightarrow \nabla^2\varphi = \partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 = 0$ (Laplace equation).

Differences & relation between ψ and φ

Stream function ψ	Potential function φ
For 2-D flow only	Any dimension (3-D also)
Exists for rotational and irrotational flow	Exists only for irrotational flow
Satisfies continuity automatically	Satisfies continuity $\Rightarrow$ Laplace equation
ψ = const $\rightarrow$ streamline	φ = const $\rightarrow$ equipotential line
$u = \partial\psi/\partial y$; $v = -\partial\psi/\partial x$	$u = -\partial\varphi/\partial x$; $v = -\partial\varphi/\partial y$
Only for incompressible flow	Can be used for compressible flow

- Relation (Cauchy-Riemann) : $\partial\varphi/\partial x = -\partial\psi/\partial y$, $\partial\varphi/\partial y = \partial\psi/\partial x$. Slope of ψ-line × slope of φ-line = $-1 \Rightarrow$ *streamlines and equipotential lines are orthogonal.*

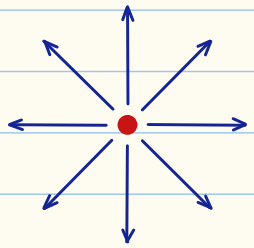
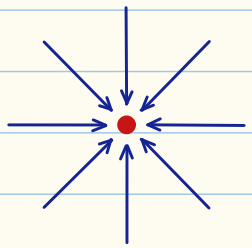
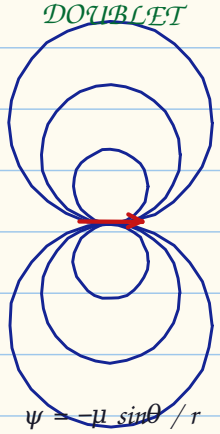
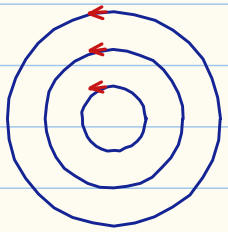
Ex 3. For $u = 2x$, $v = -2y$: $\psi = 2xy$ ($u = \partial\psi/\partial y = 2x$ □, $v = -\partial\psi/\partial x = -2y$ □) ; $\partial v/\partial x = 0 = \partial u/\partial y \Rightarrow$ irrotational, so φ exists : $\varphi = -x^2 + y^2$. (flow near a stagnation point).

Flow net

A grid of streamlines and equipotential lines intersecting at right angles forming (curvilinear) squares.

- Drawn graphically, by electrical/hele-shaw analogy or numerically. Closer lines $\Rightarrow$ higher velocity (and lower pressure).
- Uses : seepage under dams & weirs (flow through porous media), flow around bodies, through bends/contractions. Discharge $q = k H (N_f/N_d)$ for seepage.
- Limitation : only for ideal irrotational flow.

4. Basic Flows – Source, Sink, Doublet, Vortex

SOURCE	SINK	DOUBLET	FREE VORTEX
			
$\psi = (q/2\pi)\,\theta$	$\psi = -(q/2\pi)\,\theta$	$\psi = -\mu \sin\theta / r$	$v_\theta = \Gamma / 2\pi r$
q = outflow / unit depth	$v_r = -q/2\pi r$	source + sink, $d \to 0$	$\psi = -(\Gamma/2\pi) \ln r$

- ***Source*** *: flow radially outward from a point, strength q = discharge per unit depth. $v_r = q/(2\pi r)$, $v_\theta = 0$; $\varphi = -(q/2\pi) \ln r$; $\psi = (q/2\pi)\,\theta$. Sink is the negative of source.*
- ***Doublet*** *: a source and sink of equal strength brought infinitely close (distance $d \to 0$ with $q\,d = \mu$ held finite). $\psi = -\mu \sin\theta / r$. Streamlines = circles tangent to the axis.*
- ***Free (irrotational) vortex*** *: fluid circulates with $v_\theta = \Gamma/(2\pi r) \Rightarrow v\,r = const$; speed ↑ toward the centre ; pressure ↓ toward centre ; $\varphi = -\Gamma\theta/2\pi$.*
- ***Forced vortex*** *: fluid rotates like a rigid body, $v = \omega\,r$ (rotational, $\zeta = 2\omega$). Surface of liquid in a rotating cylinder : $z = \omega^2 r^2/2g$ (paraboloid).*
- *Superposition (add ψ values): uniform stream + source = Rankine half body ; source + sink = Rankine oval ; uniform stream + doublet = flow past a cylinder ; + vortex = cylinder with lift (Magnus effect).*

5. Fluid Dynamics – Forces & Euler's Equation

Forces acting on a fluid element :

- ***Surface forces*** *– act on the boundary : pressure force, shear (viscous) force, surface tension.*
- ***Body forces*** *– act on the whole mass : gravity, centrifugal, magnetic, electric.*
- *Others : inertia force (= mass × acceleration), compressibility force, turbulent force.*

Euler's equation of motion (along a streamline)

Consider a prism of length ds along a streamline : forces = pressure, weight component, no viscous force (ideal fluid). Newton's 2nd law : $p\,dA - (p + dp)\,dA - \rho g\,dA\,ds\cos\theta = \rho\,dA\,ds \cdot v\,dv/ds$.

$$dp/\rho + g\,dz + v\,dv = 0$$

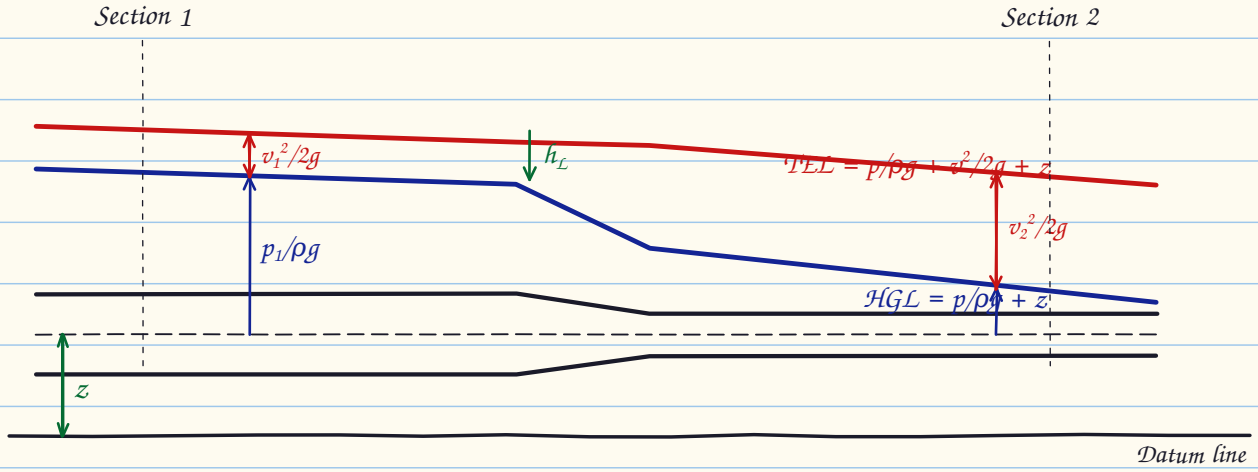
Assumptions : steady flow, along a streamline, ideal (inviscid) fluid, only gravity & pressure forces.

Bernoulli's equation

Integrate Euler's equation for an incompressible fluid (ρ = const) :

$$p/\rho g + v^2/2g + z = \text{constant}$$

- Terms (in metres of fluid) : pressure head $p/\rho g$; velocity (kinetic) head $v^2/2g$; datum (potential) head z.
- In energy per unit weight : pressure energy + kinetic energy + potential energy = const (total head H).
- Between two sections 1 and 2 with losses h_L : $p_1/\rho g + v_1^2/2g + z_1 = p_2/\rho g + v_2^2/2g + z_2 + h_L$. With pump head H_p added or turbine head H_t extracted add on the appropriate side.



- Limitations : ideal fluid, steady, incompressible, valid along a streamline (same constant for whole flow if irrotational), no heat/work transfer, correct for viscous flow only with h_L.

Applications of Bernoulli's equation

- **Venturimeter** (converging–throat–diverging) : $Q = C_d\, a_1 a_2 \sqrt{(2gh)} / \sqrt{(a_1^2 - a_2^2)}$; h = differential head = $x(S_{Hg}/S_{fluid} - 1)$ for a mercury manometer reading x.
- **Orifice meter** (cheaper, more loss, $C_d \approx 0.6$) ; **Pitot tube** : $v = C_v \sqrt{(2gh)}$ (stagnation pressure – static pressure). Also nozzles, siphons, orifices (Torricelli $v = \sqrt{(2gH)}$).

> Ex 4. Venturimeter 300 mm × 150 mm, oil-mercury differential $x = 0.2$ m, S_{oil}=1 (water) ⇒ $h = 0.2(13.6 - 1) = 2.52$ m. $a_1 = 0.0707$, $a_2 = 0.01767$ m² ; $Q = 0.98 \times (0.0707 \times 0.01767 / \sqrt{(0.0707^2 - 0.01767^2)}) \times \sqrt{(2 \times 9.81 \times 2.52)} = 0.126$ m³/s ≈ 126 l/s.

6. Momentum Equation & its Applications

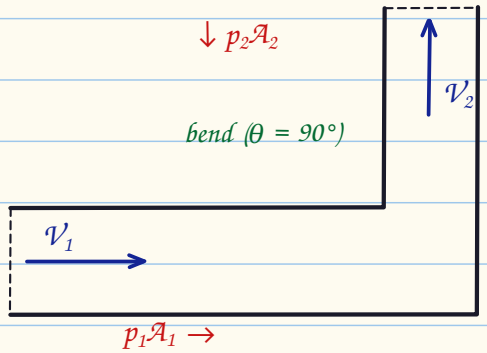
From Newton's second law, for a steady flow through a control volume :

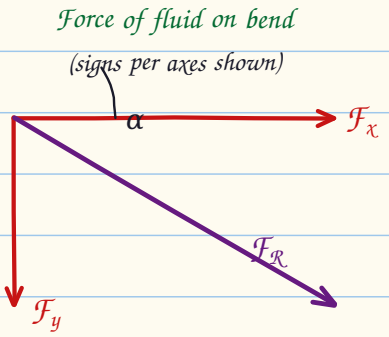
$$\Sigma F = \rho\, Q\, (V_{out} - V_{in}) \text{ (vector equation)}$$

F includes pressure forces on the end faces, body weight and the reaction of the solid boundary. Apply separately in x and y.

- Applications : force on pipe bends and reducers, force of a jet on plates and vanes, thrust on nozzles, hydraulic jump, propeller thrust, rocket motion.

Force on a pipe bend





Flow enters along x (V_1, A_1, p_1) and leaves at angle θ (V_2, A_2, p_2). Force exerted by the fluid on the bend :

$$F_x = \rho Q(V_1 - V_2 \cos\theta) + p_1 A_1 - p_2 A_2 \cos\theta$$

$$F_y = -(\rho Q V_2 + p_2 A_2)\sin\theta \quad F_R = \sqrt{(F_x^2 + F_y^2)}, \ \tan\alpha = F_y/F_x$$

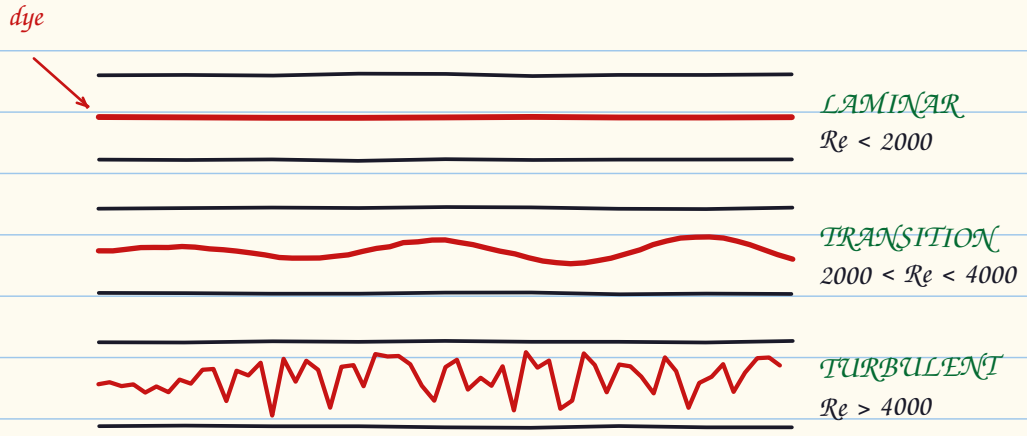
Procedure : (1) Q, V_1, V_2 from continuity ; (2) p_2 from Bernoulli ; (3) F_x, F_y ; (4) resultant and direction (the support/anchor block must resist it).

Ex 5. Horizontal 90° bend, 300 mm → 150 mm, $Q = 0.15\ m^3/s$, $p_1 = 200$ kPa. $V_1 = 2.12$, $V_2 = 8.49$ m/s ; Bernoulli ⇒ $p_2 = 200000 + 500(2.12^2 - 8.49^2) = 166.2$ kPa. $F_x = 1000(0.15)(2.12) + 200000(0.0707) = 14455$ N ; $F_y = -(1273 + 166232\times0.01767) = -4211$ N. $F_R = 15.06$ kN at $\alpha = 16.2°$ below the x-axis.

7. Closed Conduit Flow

Reynolds' experiment

Dye injected into water flowing in a glass tube at increasing velocity :



$$Re = \rho V D / \mu = V D / \nu \ \text{(ratio of inertia to viscous forces)}$$

- Pipe flow : $Re < 2000$ laminar ; 2000–4000 transition ; > 4000 turbulent. Critical velocity depends on temperature, roughness, disturbances.
- Laminar velocity profile parabolic, $V_{max} = 2\,V_{avg}$; turbulent profile flatter, $V_{avg} \approx 0.8\,V_{max}$.

Head loss – Darcy-Weisbach equation

Frictional (major) loss in a pipe of length L and diameter D :

$$h_f = f\,L\,V^2 / (2\,g\,D) = 8\,f\,L\,Q^2 / (\pi^2\,g\,D^5)$$

- f = Darcy friction factor (dimensionless). Laminar : $f = 64/Re$ (Hagen-Poiseuille : $h_f = 32\,\mu\,V\,L/(\rho\,g\,D^2)$).

- Turbulent smooth pipe : Blasius $f = 0.316\,Re^{-0.25}$ $(4000 < Re < 10^5)$. Rough pipes : $f = F(Re, \varepsilon/D)$ from Moody chart or Colebrook : $1/\sqrt{f} = -2\log(\varepsilon/3.7D + 2.51/(Re\sqrt{f}))$.
- Chezy's form : $V = C\sqrt{(m\,i)}$, $m = D/4$ (hydraulic mean depth), $i = h_f/L$. Also Fanning $f' = f/4$ $(h_f = 4 f' L V^2/2gD)$.

Ex 6. $L = 200$ m, $D = 0.1$ m, $Q = 0.02$ m³/s, $f = 0.02$: $V = 0.02/0.007854 = 2.546$ m/s ; $h_f = 0.02 \times 200 \times 2.546^2/(2\times9.81\times0.1) =$ 13.2 m.

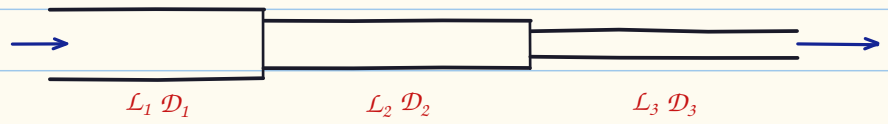
Minor losses

Losses due to change of section, direction or flow obstruction : $h_m = K V^2/2g$.

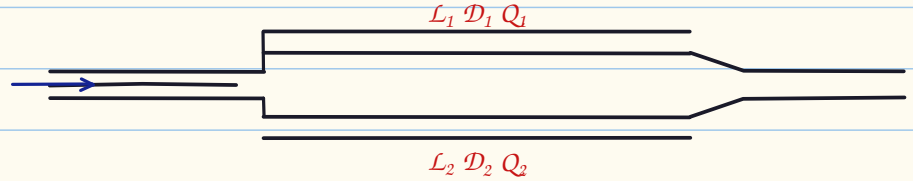
Fitting	Loss	Remark
Sudden enlargement	$h = (V_1 - V_2)^2/2g$	Borda–Carnot
Sudden contraction	$h = 0.5\,V_2^2/2g$ (approx.)	or $(1/C_c - 1)^2 V_2^2/2g$
Pipe entrance	$h = 0.5\,V^2/2g$	sharp-edged (bell-mouth 0.04)
Pipe exit to tank	$h = V^2/2g$	all KE lost
Bend / elbow 90°	$K \approx 0.4 – 0.9$	depends on R/D
Gate valve (open)	$K \approx 0.15 – 0.2$	globe valve ≈ 10
Gradual enlargement	$h = k(V_1 - V_2)^2/2g$	$k \approx 0.2 – 0.5$

- In long pipes $(L/D > 1000)$ minor losses are negligible; otherwise convert each to an equivalent length $L_e = K D / f$.

Pipes in series & parallel



SERIES : Q same
$H = h_{f1}+h_{f2}+h_{f3}$



PARALLEL : h_f same
$Q = Q_1 + Q_2 + \ldots$

- **Series** : same discharge, total loss = sum of losses : $H = \Sigma\, f L V^2/2gD$ + minor losses. Equivalent pipe (Dupuit) of length L and diameter D : $L/D^5 = L_1/D_1^5 + L_2/D_2^5 + L_3/D_3^5$.

- **Parallel** : same head loss in every branch, discharge adds : $Q = Q_1 + Q_2 + \dots$, and $h_{f1} = h_{f2} \Rightarrow Q_1/Q_2 = \sqrt{\left((D_1^5/L_1) / (D_2^5/L_2) \right)}$ (same f). Equivalent : $D^{5/2}/\sqrt{L} = \Sigma\, D_i^{5/2}/\sqrt{L_i}$.
- Pipe flowing from reservoir A to B through branching pipes (three-reservoir problem) – solved by trial for the junction piezometric head.
- Water hammer / siphon : pressure must stay above vapour pressure.

Total energy line (TEL) & Hydraulic gradient line (HGL)

- **HGL** = line joining $p/\rho g + z$ at each section (height of a piezometer column). **TEL** = $p/\rho g + v^2/2g + z$, always above HGL by $v^2/2g$.
- TEL slopes downward in direction of flow (friction); drops suddenly at fittings; rises only at a pump. For uniform pipe TEL ∥ HGL.
- If the pipe rises above the HGL, pressure there is negative (below atmosphere) – siphon action; risk of cavitation / air release.
- HGL for a pipe discharging freely to atmosphere ends at the pipe outlet centre-line.

8. Additional Solved Problems

P1 (Bernoulli). Horizontal pipe 200 mm → 100 mm, $p_1 = 150$ kPa, $V_1 = 2$ m/s (water, ideal). $V_2 = 2 \times 4 = 8$ m/s ; $p_2 = p_1 + \tfrac{1}{2}\rho(V_1^2 - V_2^2) = 150000 + 500(4 - 64) =$ **120 kPa**.

P2 (sudden enlargement). 100 mm → 200 mm, $Q = 0.03$ m³/s. $V_1 = 0.03/0.007854 = 3.82$ m/s ; $V_2 = 0.03/0.031416 = 0.955$ m/s ; $h_L = (3.82 - 0.955)^2/(2\times9.81) =$ **0.418 m**.

P3 (parallel pipes). Two pipes of equal length and same f : $D_1 = 0.2$ m, $D_2 = 0.1$ m, total $Q = 0.1$ m³/s. $Q_1/Q_2 = (D_1/D_2)^{5/2} = 2^{2.5} = 5.657 \Rightarrow Q_2 = 0.1/6.657 =$ **0.0150 m³/s** , $Q_1 =$ **0.0850 m³/s**.

P4 (orifice). Head 4 m above a 50 mm orifice, $C_d = 0.62$: $v = \sqrt{(2\times9.81\times4)} = 8.86$ m/s ; $Q = 0.62 \times 0.0019635 \times 8.86 =$ **0.0108 m³/s**.

P5 (Pitot). A Pitot-static tube in a water pipe shows a head difference of 0.15 m : $v = \sqrt{(2\,g\,h)} = \sqrt{(2\times9.81\times0.15)} =$ **1.72 m/s**.

P6 (laminar). Oil $\mu = 0.08$ Pa·s, $\rho = 900$ kg/m^3, $D = 50$ mm, $V = 0.5$ m/s : $Re = 900\times0.5\times0.05/0.08 = 281$ (laminar) ; $f = 64/281 = 0.228$; h_f per 100 m = $0.228\times100\times0.5^2/(2\times9.81\times0.05) = 5.8$ m.

Problem-solving checklist

- Draw the figure, mark sections 1 & 2, choose a datum. Write continuity first (V_2 from V_1), then Bernoulli with losses, then momentum if a force is asked.
- Pressures in Bernoulli may be all gauge or all absolute – never mix. Check Re before choosing f. Convert units (mm → m, l/s → m^3/s).
- Minor losses : find velocity in the pipe where the loss occurs (the smaller pipe for a contraction, $K V^2/2g$ with the downstream V).
- Sketch TEL and HGL after solving – TEL must fall along the flow except at a pump.

9. Quick Formula Sheet – Unit II

Topic	Formula
Continuity	$A_1V_1 = A_2V_2$; $\partial u/\partial x + \partial v/\partial y = 0$
Rotation / vorticity	$\omega = \tfrac{1}{2}(\partial v/\partial x - \partial u/\partial y)$; $\zeta = 2\omega$
ψ , φ	$u = \partial\psi/\partial y = -\partial\varphi/\partial x$; $v = -\partial\psi/\partial x = -\partial\varphi/\partial y$
Bernoulli	$p/\rho g + v^2/2g + z = const$
Momentum	$F = \rho Q(V_2 - V_1)$
Reynolds number	$Re = \rho VD/\mu$
Darcy-Weisbach	$h_f = fLV^2/2gD$; $f = 64/Re$
Minor loss	$h_m = K V^2/2g$
Series / Parallel	Σh_f ; $Q = \Sigma Q_i$

10. Important Questions

- Distinguish streamline, pathline and streakline. Derive the continuity equation in Cartesian co-ordinates.
- Define stream function and velocity potential function; give their differences and relation. Condition for irrotational flow.
- Explain source, sink, doublet and free vortex with equations. What is a flow net? State its uses.

- Derive Euler's equation and obtain Bernoulli's equation; state assumptions. Explain venturimeter with derivation.
- Derive the force exerted on a pipe bend using the momentum equation. Numerical problems.
- Explain Reynolds' experiment. Derive the Darcy-Weisbach equation. Write the Moody-chart based procedure.
- Minor losses in pipes; pipes in series and parallel; equivalent pipe; draw TEL and HGL for a pipeline.

UNIT – III

Boundary Layer Theory & Dimensional Analysis

Subject : Fluid Mechanics & Hydraulic Machines (23A03402T)

Syllabus: Boundary layer – introduction, momentum integral equation, displacement / momentum / energy thickness, separation & its control, streamlined & bluff bodies, velocity profiles. Dimensional analysis – homogeneity, non-dimensionalisation of equations, repeating-variable method, Buckingham π-theorem.

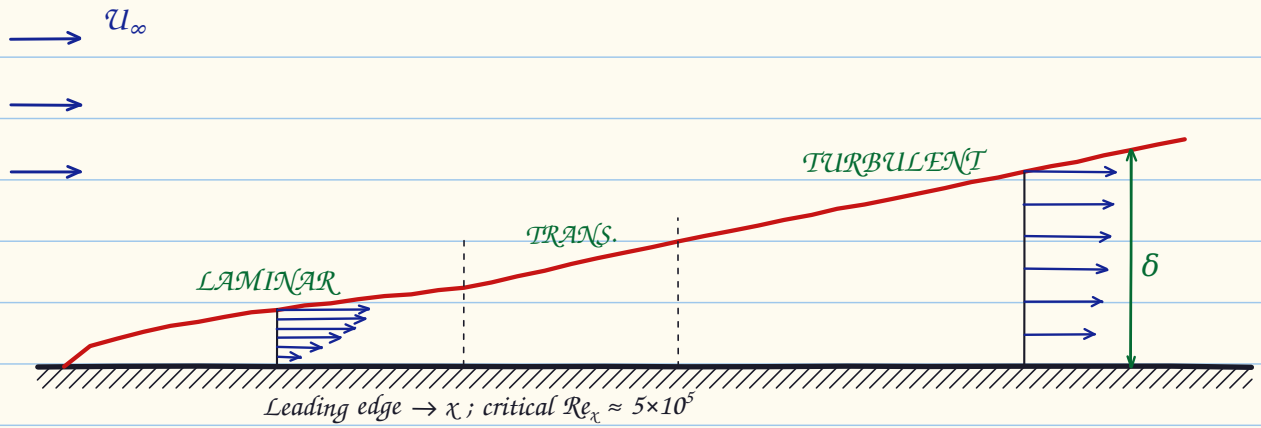
1. Boundary Layer Theory – Introduction

Ideal-fluid (potential) theory ignores viscosity and predicts zero drag (d'Alembert's paradox). Real fluids stick to a solid surface (no-slip), so velocity rises from zero at the wall to the free-stream value U_∞. Prandtl (1904) : the effect of viscosity is confined to a thin layer next to the surface – the boundary layer.

- Inside the layer : large velocity gradient $du/dy \Rightarrow$ significant shear stress $\tau = \mu\, du/dy$; viscous & inertia forces are of the same order.
- Outside the layer : fluid behaves as ideal (inviscid, irrotational) – potential flow, Bernoulli's equation can be used.
- **Boundary layer thickness δ** : distance from the wall at which $u = 0.99\, U_\infty$ (nominal thickness).

Growth of boundary layer on a flat plate

δ = thickness where $u = 0.99\ U_\infty$



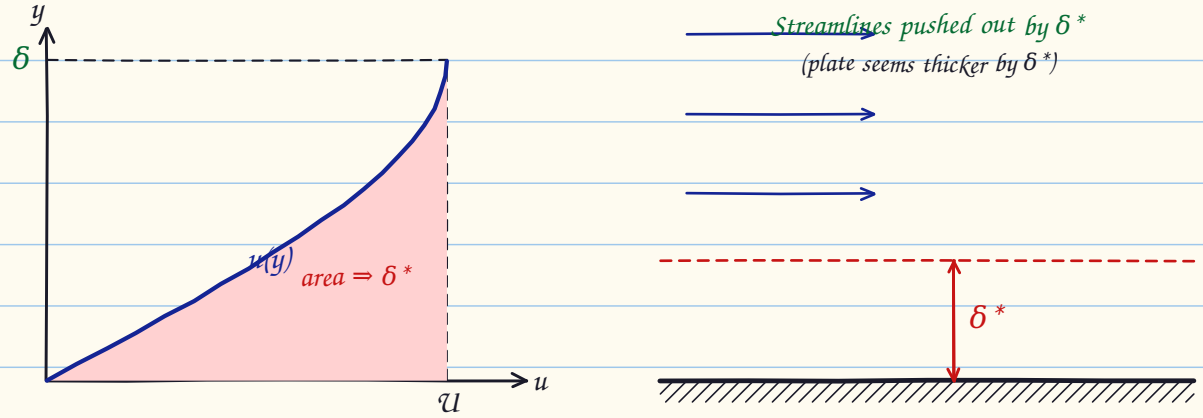
- Near the leading edge the layer is laminar (δ grows as $\sqrt{x}$). At a critical distance x_{cr} the layer turns turbulent through a transition zone.
- Local Reynolds number $Re_x = U_\infty\, x / \nu$. Critical value for flat plate $Re_{cr} \approx 5 \times 10^5$ (range $3\times10^5 – 10^6$ depending on roughness & free-stream turbulence).
- In the turbulent layer a very thin **laminar sub-layer** persists next to the wall : $\delta' = 11.6\ \nu/u_*$ where shear velocity $u_* = \sqrt{(\tau_0/\rho)}$.
- Turbulent layer is thicker, fuller velocity profile, larger wall shear stress and resists separation better than laminar.
- Boundary layers can be laminar / turbulent over smooth or rough surfaces; for pipes the layer grows until it fills the pipe (entrance length $L \approx 0.06\ Re\ D$ laminar, ≈ 25–40 D turbulent).

Characteristics of boundary layer

Item	Laminar BL	Turbulent BL
Thickness (flat plate)	$\delta = 5x/\sqrt{Re_x}$ (Blasius)	$\delta = 0.37x/Re_x^{1/5}$
Velocity profile	parabolic / cubic	1/7 power law, fuller
Local C_f	$0.664/\sqrt{Re_x}$	$0.059/Re_x^{1/5}$
Average C_D	$1.328/\sqrt{Re_L}$	$0.074/Re_L^{1/5}$ ($5\times10^5 < Re_L < 10^7$)
Separation	early (poor resistance)	later (better resistance)

2. Boundary Layer Thicknesses

Because $u \to U$ asymptotically, δ is not sharply defined. More meaningful thicknesses are integral measures :



- **Displacement thickness δ^*** : distance by which the external streamlines are displaced outward due to the reduction of flow in the layer.

$$\delta^* = \int_0^{\delta} (1 - u/U)\, dy$$

- **Momentum thickness θ** : thickness of fluid layer which, moving at U, carries the momentum lost because of the boundary layer.

$$\theta = \int_0^{\delta} (u/U)(1 - u/U)\, dy$$

- **Energy thickness δ_e** : thickness of a layer at velocity U that carries the kinetic energy deficit.

$$\delta_e = \int_0^{\delta} (u/U)(1 - u^2/U^2)\, dy$$

- Shape factor $H = \delta^*/\theta$: ≈ 2.6 laminar (Blasius), 1.3–1.4 turbulent; H rises towards 3.5-4 near separation – a flag of impending separation.

Ex 1. For $u/U = 1.5(y/\delta) - 0.5(y/\delta)^3$: $\delta^* = \delta\int(1-1.5\eta+0.5\eta^3)d\eta = \delta(1 - 0.75 + 0.125) = 3\delta/8$; $\theta = 39\delta/280 = 0.139\delta$; $H = (3/8)/(39/280) = 2.69$.

3. Momentum Integral Equation (von Kármán)

Apply the momentum theorem to a control volume of length dx and height $h > \delta$ in the boundary layer (steady, 2-D, incompressible).

- Net momentum flux out in x-direction = $\rho \int u^2\, dy$ changes by $d/dx(\rho\int u^2 dy)dx$; mass is also entrained from the free stream carrying momentum U per unit mass.

- Force on the CV = $-\tau_0\,dx$ (wall shear) – (pressure term, via Bernoulli $dp/dx = -\rho U\,dU/dx$).

Result (valid for laminar & turbulent layers) :

$$\tau_0/\rho U^2 = d\theta/dx + (2\theta + \delta^*)/U \cdot dU/dx$$

For a flat plate (U = const, $dp/dx = 0$) :

$$\tau_0 = \rho U^2\, d\theta/dx \quad C_f = \tau_0/(\tfrac{1}{2}\rho U^2) = 2\, d\theta/dx$$

Application – laminar layer with cubic profile

Assume $u/U = (3/2)\eta - (1/2)\eta^3$, $\eta = y/\delta$.

- Wall shear : $\tau_0 = \mu\,(du/dy)_0 = 3\mu U/2\delta$.
- Momentum thickness $\theta = 39\delta/280 \Rightarrow d\theta/dx = (39/280)\,d\delta/dx$.
- Equate : $3\mu U/(2\delta) = \rho U^2(39/280)\,d\delta/dx \Rightarrow \delta\,d\delta = (140/13)(\nu/U)\,dx$. Integrate ($\delta = 0$ at $x = 0$) :

$$\delta = 4.64\,x/\sqrt{Re_x} \quad C_f = 0.646/\sqrt{Re_x} \quad C_D = 1.292/\sqrt{Re_L}$$

- Exact Blasius solution gives 5.0, 0.664 and 1.328 – so the approximate method is within 4 %.
- Drag on one side of a plate of length L, width b : $F_D = C_D \cdot \tfrac{1}{2}\rho U^2 \cdot (L\,b)$.

Ex 2. Air ($\nu = 1.5\times10^{-5}$ m²/s), U = 5 m/s : at x = 1 m, $Re_x = 3.33\times10^5$ (laminar) $\Rightarrow \delta = 5(1)/\sqrt{3.33\times10^5} = 8.7$ mm.

Ex 3. Smooth plate 2 m × 1 m in water ($\nu = 10^{-6}$) at U = 0.2 m/s : $Re_L = 4\times10^5$ (laminar) ; $C_D = 1.328/\sqrt{(4\times10^5)} = 0.0021$; F_D (one side) = 0.0021 × ½(1000)(0.2)² × 2 = 0.084 N.

Turbulent layer (1/7th power-law profile)

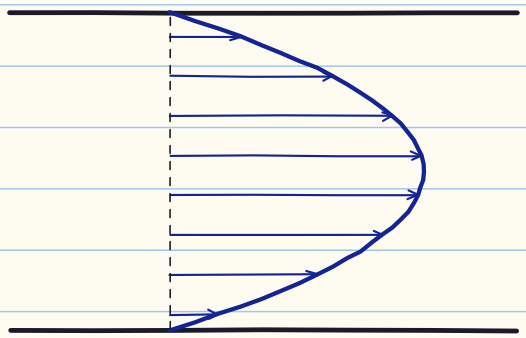
$u/U = (y/\delta)^{1/7} \Rightarrow \delta^* = \delta/8$, $\theta = 7\delta/72$. Using Blasius $\tau_0 = 0.0225\rho U^2(\nu/U\delta)^{1/4}$ in the momentum equation :

$$\delta = 0.37\,x\,Re_x^{-1/5} \quad C_f = 0.059\,Re_x^{-1/5} \quad C_D = 0.074\,Re_L^{-1/5}$$

- For plates laminar up to Re_{cr} then turbulent : $C_D = 0.074/Re_L^{1/5} - 1700/Re_L$ (Re_L up to 10^7).
- Rough surfaces : C_D depends on L/k_s – roughness raises drag in turbulent flow.

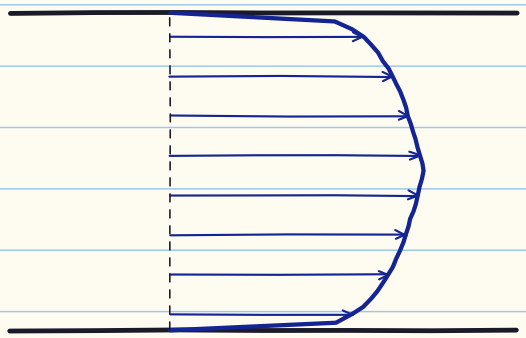
4. Velocity Profiles – Basic Concepts

LAMINAR : parabolic



$u_{max} = 2\ V_{avg}$

TURBULENT : flat (1/7 power law)



$V_{avg} \approx 0.8\ u_{max}$

- Laminar BL : $u/U = 2\eta - \eta^2$ (parabolic) or cubic (above) or sinusoidal $\sin(\pi\eta/2)$.
- Turbulent BL, wall region : universal log law (Prandtl–von Kármán)

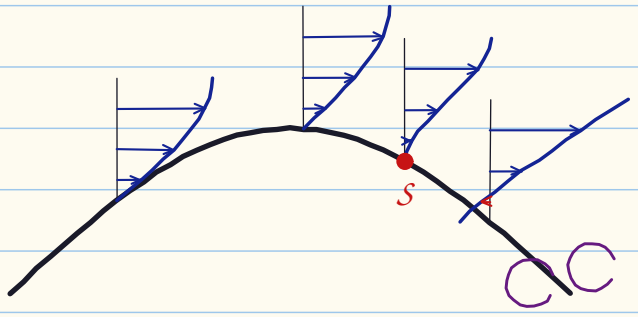
$$u/u_* = (1/\kappa)\ln(y\,u_*/\nu) + B \quad \kappa \approx 0.4\ ,\ B \approx 5.5$$

- Regions : viscous sub-layer ($y^+ < 5$, $u^+ = y^+$), buffer layer, log-law region, outer (wake) region.
- Flow in pipes : laminar parabolic $u = u_{max}[1 - (r/R)^2]$; turbulent $u/u_{max} = (y/R)^{1/n}$, $n \approx 7$ for $Re \approx 10^5$.

5. Separation of Boundary Layer

Consider flow over a curved surface (cylinder, aerofoil at high incidence, diffuser).

Flow →



Separation point S : $(\partial u/\partial y)_{y=0} = 0$

upstream of crest: $dp/dx < 0$ (favourable)

beyond crest: $dp/dx > 0$ (adverse)

reverse flow + wake ⇒ form drag

Control : suction, blowing, slots, streamlining, vortex generators

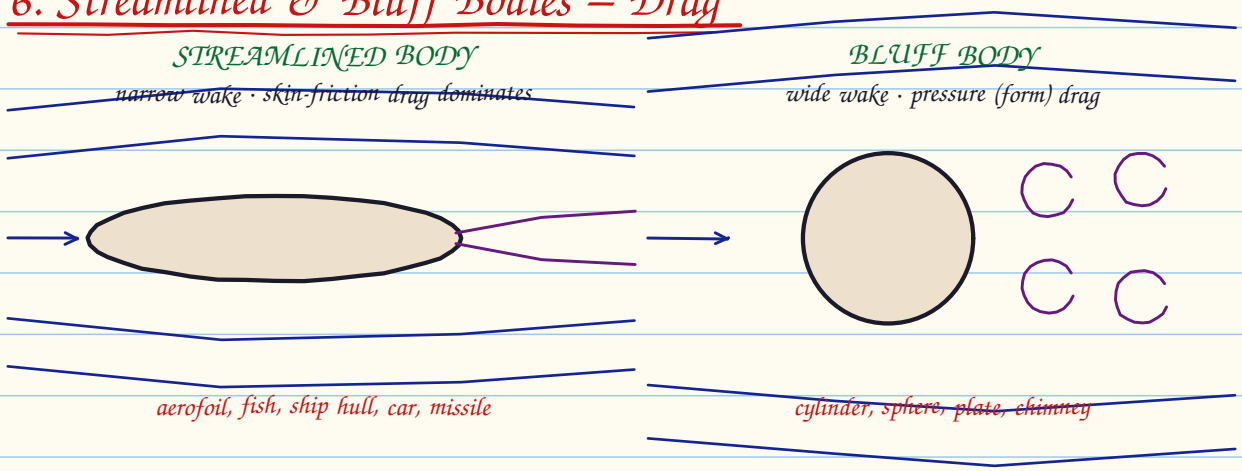
- Up to the crest the free-stream accelerates : $dp/dx < 0$ ⇒ favourable – layer thins, no separation.
- Beyond the crest the free stream decelerates : $dp/dx > 0$ ⇒ adverse pressure gradient. Fluid in the layer, already slowed by friction, loses KE and cannot climb the pressure hill.

- At the **point of separation S** : $(\partial u/\partial y)_{y=0} = 0$. Downstream, fluid near the wall flows backward → layer detaches, forms eddies/wake of low pressure.
- Consequences : large pressure (form) drag, loss of lift and stall of aerofoils, energy loss and noise, cavitation in machines, poor diffuser performance.
- Separation occurs only for adverse gradient (e.g. diverging flow); it cannot happen in converging flow. A turbulent layer separates later than laminar one – so golf balls have dimples and cricket balls are roughened to trip the layer.

Methods to control / delay separation

- Streamlining the shape (gentle taper, small divergence angle < 7° in diffusers).
- Suction of the slow layer through slots/porous walls (removes retarded fluid).
- Blowing / injection of high-energy fluid tangentially into the layer (flaps, slats on aircraft wings).
- Rotating (moving) wall in the direction of flow – Flettner rotor.
- Vortex generators / turbulators / trip wires – energise the layer to become turbulent.
- Guide vanes in bends and diffusers ; ensuring small wing angle of attack.

6. Streamlined & Bluff Bodies – Drag



Total drag = skin-friction drag (shear stress integrated over surface) + pressure (form) drag (pressure difference between front and wake).

$$F_D = C_D \cdot \tfrac{1}{2} \rho A U^2 \quad F_L = C_L \cdot \tfrac{1}{2} \rho A U^2$$

- A = projected area normal to flow (frontal) for bluff bodies ; plan area for plates/aerofoils.

- **Streamlined body** : tapered tail keeps the flow attached ; wake narrow ; drag mostly friction ; small C_D. Uses : aircraft, ships, submarines, vehicles, turbine blades, missiles.
- **Bluff body** : blunt shape, flow separates near the shoulder ; wide low-pressure wake ; drag mostly form drag. Uses : chimneys, bridge piers, buildings, poles, parachutes (large drag intended).

Body	C_D	Remark
Flat plate normal to flow	≈ 1.98	bluff
Circular cylinder ($10^3<Re<10^5$)	$\approx 1.0 – 1.2$	bluff
Sphere ($10^3<Re<2\times10^5$)	≈ 0.47	drops to 0.2 after critical Re
Streamlined strut / aerofoil	$0.04 – 0.1$	streamlined
Sphere, $Re < 1$ (Stokes)	$C_D = 24/Re$	$F_D = 3\pi\mu\, d\, U$

- Karman vortex street : alternate vortices shed behind a cylinder, Strouhal number $St = f\, d/U \approx 0.2$; may cause resonance (chimneys, cables, bridges).
- Terminal velocity : weight − buoyancy = drag ⇒ Stokes : $V_t = d^2(\rho_s - \rho)g/(18\mu)$.

Ex 4. Chimney 40 m high, 2 m dia, wind 25 m/s ($\rho = 1.2$), $C_D = 0.7$: $F_D = 0.7 \times \tfrac{1}{2}(1.2)(40\times2)(25)^2 = 21$ kN.

7. Dimensional Analysis

Method of reducing the number of variables in a physical problem by combining them into dimensionless groups.

- **Dimensional homogeneity** (Fourier) : every term of a physically correct equation has the same dimensions. e.g. Bernoulli : $p/\rho g$, $v^2/2g$, z all have dimension L.
- Uses : (i) check equations ; (ii) derive relations among variables ; (iii) plan experiments with fewer variables ; (iv) model testing – scale-up to prototype.
- Methods : Rayleigh's indicial method (≤ 4 variables) and Buckingham π-theorem (general).

Non-dimensionalisation of equations

Choose reference scales L (length), U (velocity), ρU^2 (pressure). Let $x^* = x/L$, $u^* = u/U$, $p^* = p/\rho U^2$. Navier–Stokes for incompressible flow becomes :

$$\partial u^*/\partial t^* + u^*\cdot\nabla u^* = -\nabla p^* + (1/Re)\,\nabla^2 u^* + (1/Fr^2)\,\hat{g}$$

- Each dimensionless group emerges naturally : $Re = \rho UL/\mu$, $Fr = U/\sqrt{(gL)}$ ⇒ two flows are dynamically similar if these groups are equal.

Buckingham π-theorem

If a physical phenomenon involves n variables and these contain m fundamental dimensions (M, L, T), the relation can be written as a function of (n − m) independent dimensionless π-terms : $f(\pi_1, \pi_2, \dots \pi_{n-m}) = 0$.

Method of repeating variables – steps

- 1. List all n variables (include dimensional constants such as g if needed).
- 2. Write dimensions of each in M, L, T ; find m.
- 3. Choose m repeating variables : together contain all m dimensions, no one dimensionless, no two with the same dimensions. Usual choice : one geometric (D or L), one kinematic (V), one dynamic (ρ) – never the dependent variable.
- 4. Form π-terms : each π = (non-repeating variable) × (repeating)^a (repeating)^b (repeating)^c ; solve exponents so π is dimensionless.
- 5. Express result $\pi_1 = \varphi(\pi_2, \pi_3 \dots)$; replace any π by a power or product of itself if needed (e.g. 1/Re).

Ex 5 (drag on a sphere). $F = f(D, V, \rho, \mu)$. $n = 5$, $m = 3$ ⇒ 2 π-terms. Repeating : D, V, ρ. $\pi_1 = F D^a V^b \rho^c$: M : $1 + c = 0$; T : $-2 - b = 0$; L : $1 + a + b - 3c = 0$ ⇒ $c = -1$, $b = -2$, $a = -2$ ⇒ $\pi_1 = F/(\rho V^2 D^2)$. $\pi_2 = \mu/(\rho V D) = 1/Re$. $F = \rho V^2 D^2 \varphi(Re)$ ⇒ $C_D = \varphi(Re)$.

Ex 6 (pipe pressure drop). $\Delta p = f(D, L, V, \rho, \mu, \varepsilon)$. $n = 7$, $m = 3$ ⇒ 4 π-terms. With D, V, ρ : $\Delta p/(\rho V^2) = \varphi(L/D, \varepsilon/D, Re)$; since $\Delta p \propto L$: $\Delta p = \rho V^2 (L/D) \varphi_1(Re, \varepsilon/D)$ ⇒ $h_f = f L V^2/2gD$ (Darcy-Weisbach recovered, $f = \varphi_1$).

Important dimensionless numbers

Number	Ratio	Use
Reynolds $Re = \rho VL/\mu$	inertia / viscous	pipe flow, submarines, aircraft (no free surface)
Froude $Fr = V/\sqrt{(gL)}$	inertia / gravity	open channels, ships, weirs
Euler $Eu = \Delta p/\rho V^2$	pressure / inertia	pumps, cavitation (σ)

Number	Ratio	Use
Weber $We = \rho V^2 L/\sigma$	inertia / surface tension	capillary flow, droplets
Mach $Ma = V/c$	flow / sound speed	compressible flow
Strouhal $St = fL/V$	unsteady / convective	vortex shedding

Model similarity

- **Geometric** : same shape, constant scale ratio $L_r = L_p/L_m$. **Kinematic** : velocity ratios same at corresponding points (same streamline pattern). **Dynamic** : force ratios equal (same Re, Fr…).
- Reynolds model law : $V_r = \nu_r/L_r$. Froude model law : $V_r = \sqrt{L_r}$, $Q_r = L_r^{5/2}$.

Ex 7. A ship model 1:25 is towed at 1.2 m/s. Prototype speed (Froude) : $V_p = V_m\sqrt{25} = 6$ m/s ; prototype wave drag = model drag × 25^3.

8. Lift, Stall & Magnus Effect (short notes)

- **Lift** on an aerofoil arises because the faster flow over the upper surface has lower pressure than the lower surface (Bernoulli) – equivalently, the circulation Γ gives $F_L = \rho\, U\, \Gamma\, b$ (Kutta–Joukowski).
- **Stall** : at large angle of attack ($\approx 15°$) the layer separates from the upper surface, C_L falls sharply, C_D rises. Slats, slots and flaps delay stall by re-energising the layer.
- **Magnus effect** : a spinning cylinder/sphere in a stream experiences a transverse force (circulation superposed on uniform flow) – swing of a cricket/football, Flettner rotor ship.
- **Induced drag** arises from wing-tip vortices ; winglets reduce it.
- Pipe entrance : boundary layers from the wall merge at the entrance length ; beyond this the profile is fully developed.

9. Additional Solved Problems

P1 (laminar plate). Water ($\nu = 10^{-6}$), $U = 0.4$ m/s, plate length 1 m : $Re_L = 4\times10^5$ (laminar). $\delta_L = 5(1)/\sqrt{(4\times10^5)} = 7.9$ mm ; C_f at end $= 0.664/632.5 = 0.00105 \Rightarrow \tau_0 = 0.00105 \times \tfrac{1}{2}(1000)(0.4)^2 = 0.084$ Pa ; $C_D = 1.328/632.5 = 0.0021 \Rightarrow F_D$ (one side, per metre width) $= 0.0021 \times 80 \times 1 = 0.168$ N.

P2 (turbulent plate). Air ($\rho = 1.2$, $\nu = 1.5\times10^{-5}$), $U = 30$ m/s, $L = 3$ m, $b = 1$ m : $Re_L = 6\times10^6$ (mixed layer). $C_D = 0.074/(6\times10^6)^{0.2} - 1700/6\times10^6 = 0.00326 - 0.00028 = 0.00298$. $F_D = 0.00298 \times \tfrac{1}{2}(1.2)(30)^2 \times 3 = 4.8$ N (one side).

P3 (Rayleigh's method). Time period of a simple pendulum $T = f(l, g)$. $T = K\, l^a\, g^b \Rightarrow [T] : 1 = -2b \Rightarrow b = -\tfrac{1}{2}$; $[L] : 0 = a + b \Rightarrow a = \tfrac{1}{2} \Rightarrow T = K\sqrt{(l/g)}$.

P4 (π-theorem for a pump / fan). Power $P = f(\rho, N, D)$ with $m = 3$, $n = 4 \Rightarrow 1$ π-term : $\pi = P\,\rho^a N^b D^c$; $M : 1 + a = 0$; $T : -3 - b = 0$; $L : 2 - 3a + c = 0 \Rightarrow a = -1$, $b = -3$, $c = -5 \Rightarrow P = K\rho\, N^3\, D^5$ (power coefficient).

P5 (Stokes' law). Sphere $d = 1$ mm, $\rho_s = 2650$, falling in oil ($\rho = 900$, $\mu = 0.1$ Pa·s) : $V_t = d^2(\rho_s - \rho)g/(18\mu) = 10^{-6}(1750)(9.81)/(1.8) = 9.5$ mm/s ; check $Re = 900(0.00954)(0.001)/0.1 = 0.086 < 1$ ✓ (Stokes valid).

P6 (Froude model). Spillway model scale 1 : 20, model discharge 0.02 m^3/s : $Q_r = L_r^{5/2} = 20^{2.5} = 1789 \Rightarrow Q_p = 0.02 \times 1789 = 35.8\ m^3/s$.

Common mistakes to avoid

- Taking δ as the point where $u = U$ exactly – use 0.99 U. Using laminar formulas when $Re_L > 5\times10^5$.
- Selecting repeating variables that are not independent (e.g. V, L and Q) or including the dependent variable among them.
- Forgetting that separation needs an *adverse* gradient – accelerating flow never separates.
- Mixing up area : bluff body → frontal area ; plate/aerofoil → plan area.

10. Short Answers – Quick Revision

- **Why does the boundary layer thicken downstream?** Viscous effect diffuses further from the wall as more fluid is slowed ; $\delta \propto \sqrt{x}$ (laminar).
- **Why is wall shear greater in a turbulent layer?** Fuller profile gives a steeper gradient $(du/dy)_{wall}$; momentum exchange by eddies.

- *Why use dimples on a golf ball?* They trip the layer to turbulence, delaying separation, shrinking the wake and cutting form drag by ~ 50 %.
- *What is d'Alembert's paradox?* Potential flow past a body predicts zero drag – resolved by the boundary layer & wake.
- *Is displacement thickness always smaller than δ?* Yes ; $\delta^* < \theta \cdot H$ and $\delta^* < \delta$ (for turbulent $\delta^* \approx \delta/8$, laminar $\approx 0.375\delta$).
- *What if two numbers (Re and Fr) must be matched?* Usually impossible with same fluid – choose the dominant force (Fr for free-surface, Re for pressure-driven flows).
- *Number of π-terms for 6 variables, 3 dimensions?* $6 - 3 = 3$.
- *When is the 1/7th power law invalid?* $Re > 10^7$ (use log law), near the wall ($y^+ < 30$) and in separated flow.

11. Quick Formula Sheet – Unit III

Item	Formula
Thicknesses	$\delta^* = \int(1-u/U)dy$; $\theta = \int(u/U)(1-u/U)dy$
Momentum integral	$\tau_0 = \rho U^2 \, d\theta/dx$
Laminar flat plate	$\delta = 5x/\sqrt{Re_x}$; $C_D = 1.328/\sqrt{Re_L}$
Turbulent flat plate	$\delta = 0.37x/Re_x^{0.2}$; $C_D = 0.074/Re_L^{0.2}$
Drag, lift	$F = C \cdot \frac{1}{2}\rho A U^2$
Separation	$(\partial u/\partial y)_{y=0} = 0$; $dp/dx > 0$
π-theorem	number of $\pi = n - m$
Froude / Reynolds model	$V_r = \sqrt{L_r}$; $V_r = \nu_r/L_r$

12. Important Questions

- Define boundary layer. Explain its growth over a flat plate with a neat sketch; laminar sub-layer.
- Define displacement, momentum and energy thickness and derive expressions. Find H for a given profile.
- Derive von Kármán momentum integral equation. Use it to obtain δ, C_f, C_D for a laminar layer with cubic profile.

- What is separation? Explain with velocity profiles; state methods of control of separation.
- Distinguish streamlined and bluff bodies with examples; define drag & lift coefficients.
- State and prove Buckingham π-theorem steps; solve a problem by repeating-variable method. State uses of dimensionless numbers.

UNIT – IV

Impact of Jets & Hydraulic Turbines

Subject : Fluid Mechanics & Hydraulic Machines (23A03402T)

Syllabus: Hydrodynamic force of jets on stationary & moving flat, inclined and curved vanes, jet striking centrally and at tip, velocity diagrams, work done & efficiency, radial vanes. Hydraulic turbines – classification, impulse & reaction, Pelton, Francis, Kaplan, working proportions, work done, efficiencies, hydraulic design, draft tube.

1. Impact of Jets – Basics

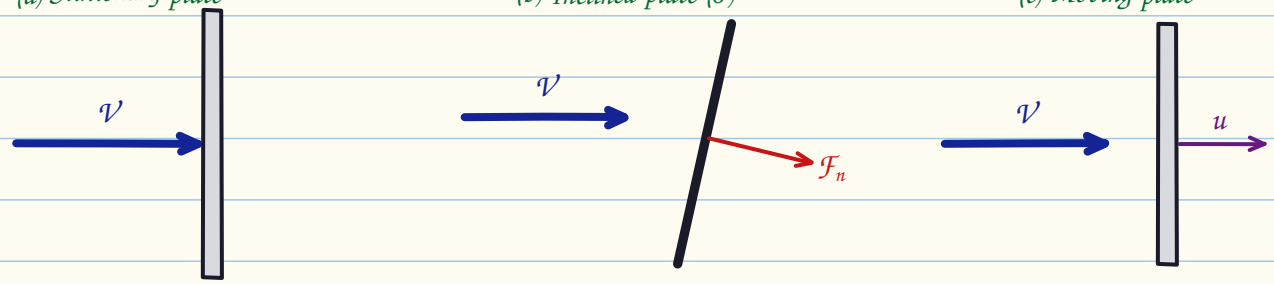
A jet of water leaving a nozzle at atmospheric pressure has only kinetic energy. When it strikes a plate or vane its momentum changes and a force is exerted. From the momentum equation :

$$F = \rho\, Q\,(V_{in} - V_{out}) \text{ (in the direction considered)}, \quad Q = a\,V$$

- a = cross-section area of jet, V = jet velocity, u = velocity of the vane in the direction of the jet.
- Mass of water striking per second : stationary vane → $\rho a V$; single moving vane → $\rho a(V - u)$; series of vanes (wheel) → $\rho a V$ (all the water issuing from the nozzle is used because a new vane always takes the place of the old one).
- Force on the plate = rate of change of momentum ; Work done = $F \times u$; Jet power = $\frac{1}{2}\, \rho\, a\, V^3$; Efficiency η = work done/ jet power.

(A) Flat plates

(a) Stationary plate (b) Inclined plate (θ) (c) Moving plate



$F = \rho\, a\, V^2$ $F_n = \rho\, a\, V^2 \sin\theta$ $F = \rho\, a\, (V - u)^2$

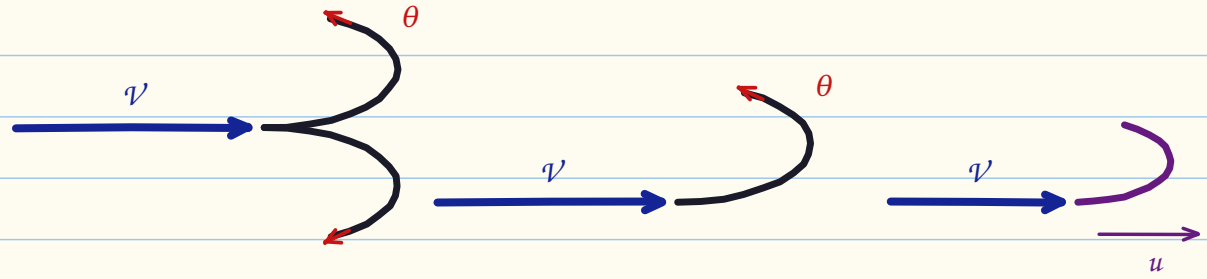
- ***Stationary plate, jet normal*** : $F = \rho\, a\, V^2$ (water leaves along the plate – no component in jet direction).
- ***Stationary inclined plate*** (plate inclined θ to jet axis, smooth) : normal force $F_n = \rho\, a\, V^2 \sin\theta$; component along jet $F_x = \rho\, a\, V^2 \sin^2\theta$; perpendicular to jet $F_y = \rho\, a\, V^2 \sin\theta \cos\theta$. The jet divides into two streams on the plate : $Q_1 = \frac{1}{2}Q(1 + \cos\theta)$, $Q_2 = \frac{1}{2}Q(1 - \cos\theta)$.
- ***Hinged plate*** : plate hinged at top, jet strikes at its centre – equate moments : $F_n \times (L/2) = W \times (L/2) \sin\alpha \Rightarrow$ angle α of swing of the plate.
- ***Moving plate, normal*** (single) : $F = \rho\, a\, (V - u)^2$; work/sec $= F\, u$; $\eta = 2u(V - u)^2/V^3$; maximum $\eta = 8/27 = 29.6\ \%$ at $u = V/3$.
- ***Series of plates (wheel)*** : $F = \rho\, a\, V\, (V - u)$; work done $= \rho\, a\, V\, (V - u)\, u$;

$$\eta = 2u(V - u)/V^2 \quad \eta_{max} = 50\ \%\ \text{at}\ u = V/2$$

Ex 1. Jet 50 mm dia, V = 20 m/s strikes a stationary flat plate normal : $F = 1000 \times (\pi/4)(0.05)^2 \times 20^2 = 1000 \times 0.0019635 \times 400 = 785$ N. If the plate moves at 5 m/s along the jet : $F = 1000(0.0019635)(15)^2 = 442$ N, work $= 442 \times 5 = 2.21$ kW.

(B) Curved vanes

(a) Stationary, jet at centre (b) Jet at tip (tangential) (c) Moving vane



$F = \rho a V^2(1 + \cos\theta)$ $F_x = \rho a V^2(1 + \cos\theta)$ $F_x = \rho a(V-u)^2(1+\cos\theta)$

Curved vanes turn the jet through a large angle, so the momentum change is greater. Let θ = angle between the jet leaving the tip and the direction of motion (outlet angle). A smooth vane has $V_{r2} = V_{r1}$.

- ***Stationary symmetrical vane, jet at centre*** : jet splits equally ; force in jet direction : $F_x = \rho\ a\ V^2(1 + \cos\theta)$. For a semi-circular vane ($\theta = 0$) $F = 2\rho a V^2$ – twice that on a flat plate.
- ***Stationary unsymmetrical vane, jet at one tip (tangential)*** : $F_x = \rho\ a\ V^2(\cos\theta + \cos\varphi)$, $F_y = \rho\ a\ V^2(\sin\theta - \sin\varphi)$ where θ , φ are inlet and outlet tip angles. Resultant $F = \sqrt{(F_x^2 + F_y^2)}$.
- ***Moving curved vane (single)*** : $F_x = \rho\ a\ (V - u)^2(1 + \cos\theta)$.
- ***Series of curved vanes (wheel, e.g. Pelton)*** : mass $\rho a V$ per second ; relative velocity at inlet $V_{r1} = V - u$; at outlet $V_{r2} = V_{r1}$ (smooth) ; whirl at outlet $V_{w2} = V_{r2}\cos\varphi - u$ (φ = outlet vane angle with the –ve direction of u).

$$F_x = \rho\ a\ V\ (V - u)(1 + \cos\varphi)\quad Work = F_x\ u$$

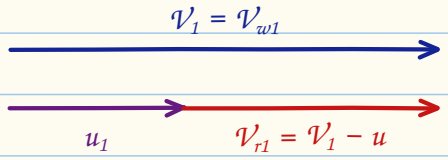
$$\eta = 2u\ (V - u)(1 + \cos\varphi)\ /\ V^2\quad \text{maximum at } u = V/2 : \eta_{max} = (1 + \cos\varphi)/2$$

- For $\varphi = 0$ (semicircular vane, ideal) $\eta_{max} = 100\ \%$ – all the KE is converted. In practice $\varphi \approx 15° \Rightarrow \eta_{max} \approx 98\ \%$.

Ex 2. Jet 75 mm dia, $V = 30$ m/s strikes the centre of a stationary symmetrical vane that deflects it through 150° ($\theta = 30°$). $F = \rho a V^2(1 + \cos 30°) = 1000 \times 0.004418 \times 900 \times 1.866 =$ 7420 N.

Velocity diagrams – work done

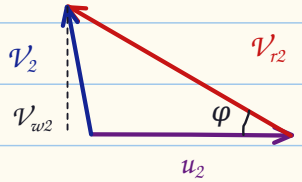
INLET triangle ($\alpha = 0$)



Pelton / moving vane

$V_{f1} = 0$

OUTLET triangle



$V_{w2} = V_{r2}\cos\varphi - u_2$

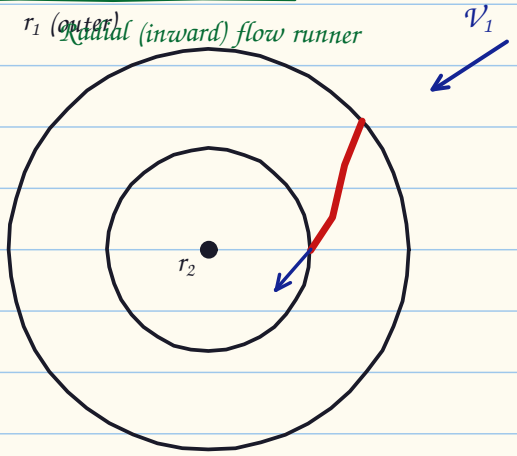
For a general moving vane (inlet angle α, vane angles θ and φ) the force in the direction of motion is $F_x = \rho\, Q\,(V_{w1} \pm V_{w2})$; '+' when the outlet whirl V_{w2} is in the opposite direction to u (vane angle $\varphi < 90°$), '–' when in the same direction.

Work done per second = $\rho\, Q\,(V_{w1} \pm V_{w2})\, u$ Work done per kg = $(V_{w1} \pm V_{w2})\, u$

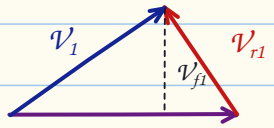
Jet striking at the tip vs centrally

- Centre : symmetrical vane, jet splits, side thrust cancels – only axial force ; used in Pelton buckets (splitter at the centre).
- Tip : unsymmetrical vane, flow enters tangentially – there is side force F_y; ensures shock-free entry when vane angle = jet direction.

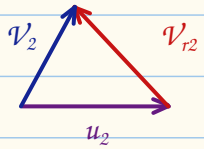
Flow over radial vanes



INLET (α , θ)



OUTLET (β , φ)



$W = \rho Q(V_{w1}u_1 \pm V_{w2}u_2)$

In a wheel with radial vanes (inlet radius r_1, outlet r_2) the velocities are tangential ($u = \omega\, r$), so the angular momentum equation gives torque :

$T = \rho\, Q\,(\, V_{w1}\, r_1 \pm V_{w2}\, r_2\,)$ Work done = $T\,\omega = \rho\, Q\,(V_{w1}u_1 \pm V_{w2}u_2)$

- Radial inward flow (turbine) : water enters at outer radius r_1, leaves at inner r_2. Radial outward flow (centrifugal pump) is the reverse and the work is added to the fluid.
- Hydraulic efficiency $\eta_h = (V_{w1}u_1 \pm V_{w2}u_2) / (g\,H)$. This is the **Euler turbine equation** – basis of all turbomachines.

Ex 3. Radial vane wheel : $u_1 = 25$ m/s, $V_{w1} = 18$ m/s, $u_2 = 12$ m/s, $V_{w2} = 0$ (radial discharge), $Q = 0.5\ m^3/s \Rightarrow$ work $= 1000 \times 0.5 \times (18 \times 25) =$ 225 kW.

2. Hydraulic Turbines – Classification

Turbines convert the hydraulic energy of water (head) into mechanical energy of a rotating shaft, which drives a generator. Gross head H_g = head race level – tail race level ; net (effective) head $H = H_g - h_f$ (friction loss in penstock).

Basis	Types	Example
Energy at inlet	Impulse (only KE ; atmospheric) / Reaction (KE + pressure)	Pelton / Francis, Kaplan
Direction of flow	Tangential / radial (inward) / axial / mixed	Pelton / Francis (old) / Kaplan / modern Francis
Head	High (> 250 m) / medium (60–250 m) / low (< 60 m)	Pelton / Francis / Kaplan, propeller
Specific speed N_s	Low (< 60) / medium (60–400) / high (> 400)	Pelton / Francis / Kaplan
Shaft position	Horizontal / vertical	Pelton often H ; Francis, Kaplan V
Discharge	Low / medium / high	Pelton / Francis / Kaplan

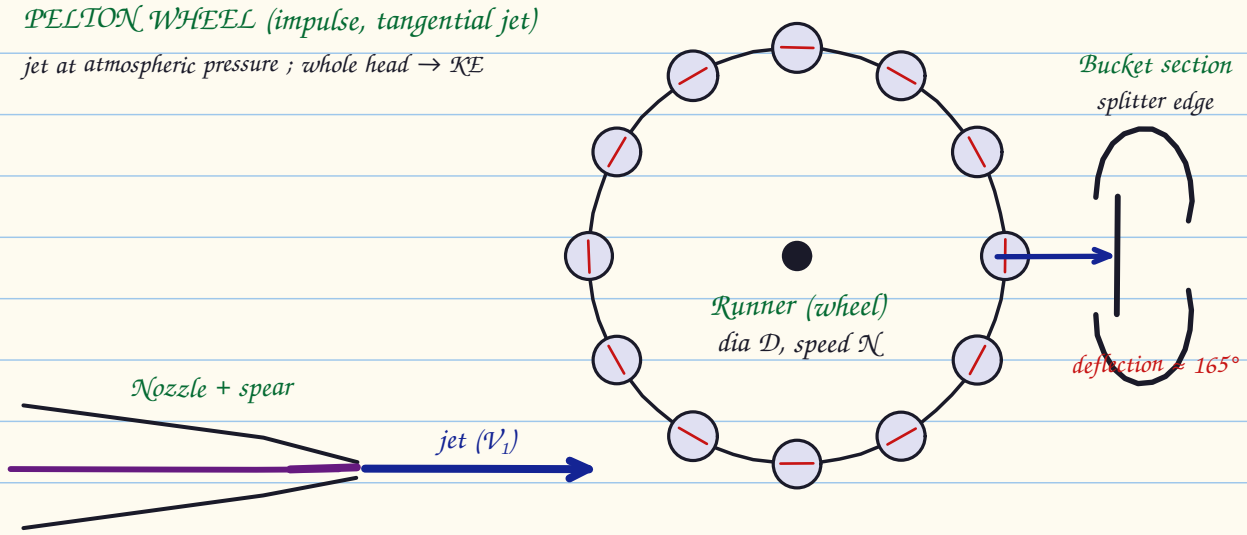
Efficiencies & power

Water power $WP = \rho\, g\, Q\, H$ Runner power $RP = \rho Q(V_{w1}u_1 \pm V_{w2}u_2)$ Shaft power SP

Efficiency	Definition	Typical value
Hydraulic η_h	RP / WP = runner power / water power	85 – 94 %
Mechanical η_m	SP / RP (bearing, windage loss)	95 – 99 %

Efficiency	Definition	Typical value
Overall η_o	$SP / WP = \eta_h \times \eta_m$	80 – 92 %
Volumetric η_v	$(Q - q_{leak})/Q$	97 – 99 %

3. Pelton Wheel (Impulse Turbine)



Working : water from the penstock reaches the nozzle, whose spear controls the jet diameter. The whole head is converted to KE ; the jet at atmospheric pressure strikes the double-cup buckets at the splitter, is turned through about 165° and falls into the tail race. Runner rotates in air – casing only prevents splashing. A brake nozzle stops the wheel quickly.

Parts

- **Nozzle and spear** (needle) – regulates discharge ; **deflector** – diverts the jet from the buckets on load rejection to avoid water hammer ; **buckets** – double hemispherical cups with a central splitter ; **casing** ; **penstock** ; **runner disc.**

Velocities, work done and efficiency

- Jet velocity $V_1 = C_v \sqrt{(2gH)}$ ($C_v \approx 0.97$–0.99). Bucket velocity $u = \varphi_s \sqrt{(2gH)}$ where speed ratio $\varphi_s \approx 0.43 - 0.48$.
- $V_{w1} = V_1$; $V_{r1} = V_1 - u$; $V_{r2} = k\, V_{r1}$ ($k \approx 0.9$–0.95 for friction) ; $V_{w2} = V_{r2} \cos\varphi - u$; $\varphi = 180° -$ bucket angle $\approx 15°$.

$$\text{Work done / s} = \rho\, Q\, (V_1 - u)(1 + k\cos\varphi)\, u$$

$$\eta_h = 2\,(V_1 - u)(1 + k\cos\varphi)\, u / V_1^2 \quad \eta_{max} \text{ when } u = V_1/2 : (1 + k\cos\varphi)/2$$

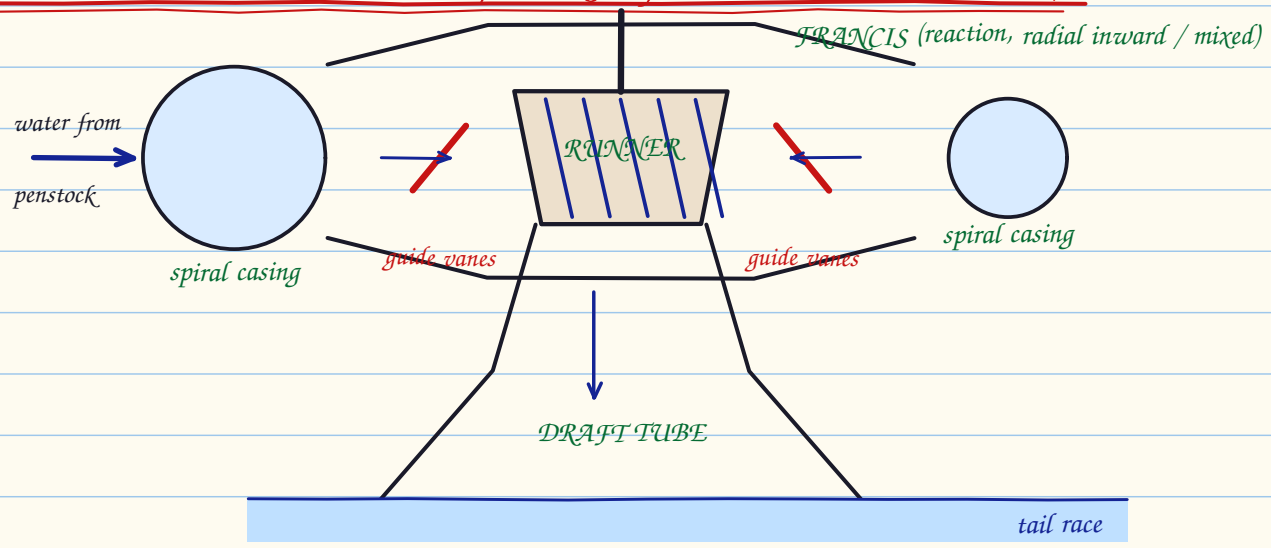
- Power available at nozzle = $\rho g Q H$; $N = 60u/(\pi D)$.

Hydraulic design (working proportions)

- Jet diameter $d = \sqrt{(4Q/(\pi V_1))}$. **Jet ratio** $m = D/d = 10 – 20$ (12 normally) ; number of buckets $z = 15 + D/(2d) = 15 + m/2$.
- Bucket : width ≈ 5d, depth ≈ 1.2d ; deflection 165°. Maximum jets per wheel (vertical shaft) : 6. Specific speed 10 – 60 for one jet (increases as √(number of jets)).
- Governing : spear moves in/out to change Q while V stays the same ⇒ efficiency remains almost constant over wide load range.

Ex 4. Pelton wheel : $H = 300$ m, $D = 1.5$ m, jet $d = 100$ mm, $C_v = 0.98$, $\varphi_s = 0.45$, $k = 1$, $\varphi = 15°$. $V_1 = 0.98\sqrt{(2\times9.81\times300)} = 75.2$ m/s ; $Q = 0.007854 \times 75.2 = 0.59$ m^3/s ; $u = 0.45 \times 76.7 = 34.5$ m/s ⇒ $N = 60u/\pi D =$ 440 rpm. $V_{r1} = 40.7$; $V_{w2} = 40.7\cos15° – 34.5 = 4.8$ m/s. Power = $1000 \times 0.59 \times (75.2 + 4.8) \times 34.5$ = 1630 kW ; WP = 1738 kW ⇒ $\eta_h =$ 93.8 %.

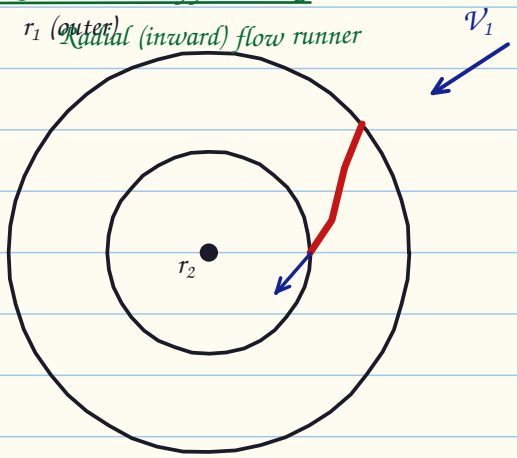
4. Francis Turbine (Radial / Mixed-flow Reaction Turbine)

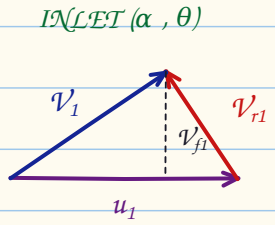


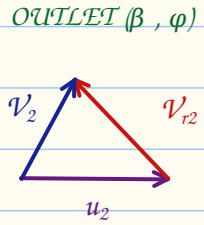
Working : water from the penstock enters the spiral (scroll) casing, whose cross-section decreases so that velocity stays uniform around the periphery. It passes through stay vanes and adjustable **guide vanes (wicket gates)** which give the flow the correct angle α, and enters the runner radially, flowing inward and leaving axially into the **draft tube.** The runner is completely filled with water, at pressure above atmosphere at inlet and below at exit – both pressure and kinetic energy are converted.

- Parts : spiral casing, stay ring, guide vanes (governor linkage), runner (16–24 blades, curved backward), draft tube, shaft.

Work done & efficiency







$W = \rho Q(V_{w1}u_1 \pm V_{w2}u_2)$

$$\text{Work/s} = \rho\, Q\,(V_{w1}\,u_1 \pm V_{w2}\,u_2)\,\eta_h = (V_{w1}u_1 \pm V_{w2}u_2)\,/\,(g\,H)$$

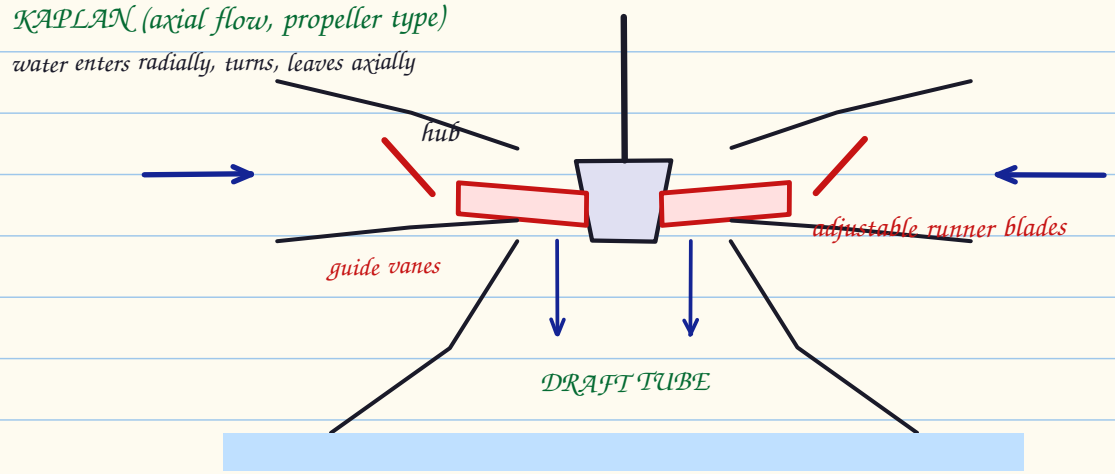
- For radial discharge ($V_{w2} = 0$, exit blade angle ~ 90°) : $\eta_h = V_{w1}\,u_1 / (g\,H)$. Discharge $Q = \pi\, D_1\, B_1\, V_{f1} = \pi\, D_2\, B_2\, V_{f2}$ (reduce by vane thickness factor $k_t \approx 0.95$).
- **Degree of reaction** R = (pressure energy change in runner) / (total energy change) $= 1 - (V_1^2 - V_2^2)/(2\,g\,E)$, Francis $R \approx 0.4$–0.6 ; impulse $R = 0$; Kaplan $R \approx 0.5$-0.7.

Working proportions & design

- Speed ratio $K_u = u_1/\sqrt{(2gH)} = 0.6 - 0.9$; flow ratio $K_f = V_{f1}/\sqrt{(2gH)} = 0.15 - 0.30$. Breadth ratio $n = B_1/D_1 = 0.10 - 0.45$; ratio $D_1/D_2 \approx 1.5$-2 ; guide vane angle $\alpha = 15°$–$30°$; runner vane angle at inlet $\theta = 60°$–$120°$.
- Governing by changing the opening of the guide vanes ; $N = 60u_1/(\pi D_1)$.

Ex 5. Francis : $D_1 = 1.2$ m, $N = 300$ rpm, $B_1 = 0.15$ m, $\alpha = 20°$, $V_{f1} = 6$ m/s, $H = 40$ m, radial discharge. $u_1 = \pi(1.2)(300)/60 = 18.85$ m/s ; $V_{w1} = 6/\tan 20° = 16.5$ m/s ; $\eta_h = 16.5\times18.85/(9.81\times40) = 79\ \%$; $Q = \pi\, D_1\, B_1\, V_{f1} = 3.39\ m^3/s$; power $= 1000 \times 3.39 \times 16.5 \times 18.85 = 1.05\ MW$.

5. Kaplan Turbine (Axial-flow Reaction Turbine)



- A propeller turbine with adjustable runner blades (and adjustable guide vanes → double regulation). Water enters through the spiral casing and guide vanes, turns through 90° in the vaneless space and flows axially through the runner. Used for low head (5 – 60 m) and very large discharge. Propeller turbine = fixed blades (good only near design load).
- Parts : scroll casing, guide vanes, hub (boss) carrying 3 – 8 blades, draft tube. Blade angle is adjusted by a servo mechanism in the hub as load varies, keeping efficiency high from 30 % to 100 % load.
- Since flow is axial, $u_1 = u_2 = \pi D_o N/60$ (at blade outer rim) ; D_o = runner dia, D_b = boss dia.

$$Q = (\pi/4)(D_o^2 - D_b^2)\, V_{f1} \quad \text{Work/s} = \rho\, Q\,(V_{w1} \pm V_{w2})\, u \quad \eta_h = (V_{w1} \pm V_{w2})\, u/gH$$

- Proportions : hub ratio D_b/D_o = 0.35 – 0.6 ; speed ratio 1.4 – 2.0 ; flow ratio 0.5 – 0.7 ; specific speed 300 – 1000.

Ex 6. Kaplan : H = 20 m, D_o = 3 m, D_b = 1.2 m, K_f = 0.5, η_o = 0.88. $V_{f1} = 0.5\sqrt{(2\times9.81\times20)} = 9.9$ m/s ; $Q = (\pi/4)(9 - 1.44)(9.9) = 58.8$ m³/s ; $P = \eta_o \rho g Q H = 0.88 \times 9810 \times 58.8 \times 20 =$ 10.2 MW.

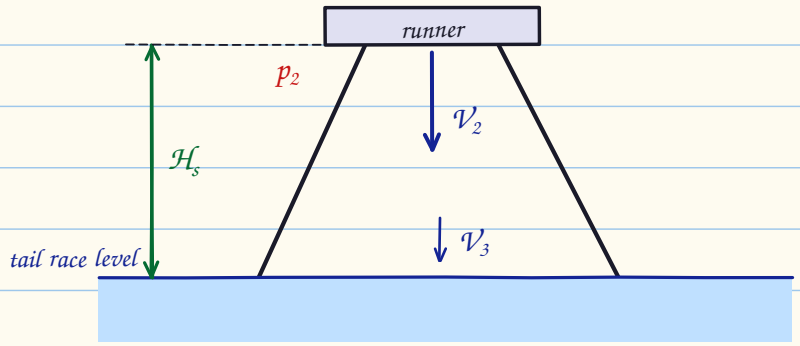
Comparison of the three turbines

Item	Pelton	Francis	Kaplan
Type	Impulse	Reaction	Reaction
Head	High > 250 m	Medium 60–250 m	Low < 60 m
Discharge	Low	Medium	High
Flow direction	Tangential	Radial → axial	Axial

Item	Pelton	Francis	Kaplan
N_s	10 – 60	60 – 400	300 – 1000
Regulation	Spear + deflector	Guide vanes	Guide vanes + blades
Draft tube	Not needed	Needed	Needed
Efficiency curve	flat at part load	falls at part load	flat (double regulation)

6. Draft Tube

A gradually expanding conduit connecting the runner exit to the tail race, completely submerged at its outlet (used in reaction turbines).



$\eta_d = [(V_2^2 - V_3^2)/2g - h_f] / (V_2^2/2g)$

$p_2/\rho g = p_{atm}/\rho g - H_s - \eta_d V_2^2/2g$

(p_2 below atmospheric ⇒ extra head)

conical tube : divergence 7–8° , $\eta_d \approx 90$ %

Functions

- Permits the turbine to be set above tail-water level (for easy inspection/maintenance) without losing the head H_s – because the pressure at runner exit is below atmospheric (suction).
- Converts the large kinetic energy leaving the runner ($V_2^2/2g$) into pressure energy by gradual expansion – recovered as additional effective head ; reduces the exit loss to $V_3^2/2g$.
- Keeps the discharge from the runner under water seal and prevents air from entering.

Theory & efficiency

Bernoulli between runner exit 2 and tail-race level 3 : $p_2/\rho g + V_2^2/2g + H_s = p_{atm}/\rho g + V_3^2/2g + h_f$.

$$\eta_d = [\,(V_2^2 - V_3^2)/2g - h_f\,] / (\,V_2^2/2g\,) \quad p_2/\rho g = p_{atm}/\rho g - H_s - \eta_d\, V_2^2/2g$$

- Larger η_d ⇒ lower pressure at runner exit ⇒ more head utilised – but pressure must stay above the vapour pressure (cavitation limit – see Unit V).

Type	Description	η_d
Conical (straight divergent)	cone angle ≤ 8–10° ; H_s < 8 m	≈ 90 %
Simple elbow	elbow with constant / expanding section	60 – 65 %
Moody spreading (hydracone)	conical with central bell	up to 85 %
Elbow with varying section	bend + rectangular outlet (most common for Francis/Kaplan)	≈ 85 %

- Divergence angle must be ≤ 10° – beyond this the flow separates from the wall (adverse pressure gradient) and the tube loses its recovery.

Ex 7. Draft tube inlet dia 1.0 m, outlet 1.6 m, $Q = 3\ m^3/s$: $V_2 = 3.82$, $V_3 = 1.49\ m/s$; if $h_f = 0.1\ m$: $\eta_d = [(14.6 - 2.22)/19.62 - 0.1]/(14.6/19.62) = (0.631 - 0.1)/0.744 = 71\ \%$.

7. Quick Formula Sheet – Unit IV

Item	Formula
Flat plate normal	$F = \rho a V^2$; moving: $\rho a(V-u)^2$; wheel: $\rho a V(V-u)$
Inclined plate	$F_n = \rho a V^2 \sin\theta$
Curved vane (wheel)	$F_x = \rho a V(V-u)(1+\cos\varphi)$
Efficiency (wheel)	$2u(V-u)(1+\cos\varphi)/V^2$; max at $u = V/2$
Euler equation	$W = \rho Q(V_{w1}u_1 \pm V_{w2}u_2)$
Pelton	$V_1 = C_v\sqrt{(2gH)}$; $u = 0.45\sqrt{(2gH)}$; $m = D/d$
Francis	$\eta_h = V_{w1}u_1/gH$; $Q = \pi D_1 B_1 V_{f1}$
Kaplan	$Q = (\pi/4)(D_o^2 - D_b^2)V_f$
Efficiencies	$\eta_o = \eta_h\eta_m = SP/WP$

8. Important Questions

- Derive the force exerted by a jet on a (i) stationary (ii) moving flat plate; find the condition for maximum efficiency of a series of plates.
- Obtain expressions for work done and efficiency of a jet on a series of curved vanes; show $\eta_{max} = 50\ \%$ for flat and $(1+\cos\varphi)/2$ for curved vanes.

- Draw velocity triangles for a Pelton bucket and derive efficiency; explain governing and working proportions.
- Classify hydraulic turbines. Define different efficiencies of a turbine.
- Explain with neat sketch the working of Francis turbine; derive the work done and efficiency; define degree of reaction.
- Explain Kaplan turbine and compare it with propeller and Francis turbines.
- What is a draft tube? Explain its functions, types and efficiency; derive the pressure at runner exit.

UNIT – V

Turbine Performance, Hydraulic Systems & Pumps

Subject : Fluid Mechanics & Hydraulic Machines (23A03402T)

> Syllabus: Performance of hydraulic turbines (similarity, unit & specific quantities, characteristic curves, governing, selection, cavitation, surge tank, water hammer) ; hydraulic ram, lift, coupling ; fluidics ; centrifugal pumps ; reciprocating pumps.

1. Performance of Hydraulic Turbines

Geometric similarity & model testing

Model and prototype are geometrically similar (same shape, all linear dimensions in a fixed ratio) and operate at dynamically similar conditions (same velocity triangle shapes / same Euler, Reynolds numbers). Turbine models (1:5 – 1:30) are tested to predict the prototype performance.

- Scale ratio $L_r = D_p/D_m$. Similarity laws (same fluid, similar flow) :

$$gH/(N^2D^2) = \text{const}\quad Q/(N D^3) = \text{const}\quad P/(\rho N^3 D^5) = \text{const}$$

- Hence for the same turbine (D fixed) : $N \propto \sqrt{H}$, $Q \propto \sqrt{H}$, $P \propto H^{3/2}$.
- Efficiency scale effect : prototype efficiency is higher than the model's (Moody formula : $(1-\eta_p)/(1-\eta_m) = (D_m/D_p)^{1/5}$).

Unit quantities (for a head of 1 m, same turbine)

$$N_u = N/\sqrt{H}\quad Q_u = Q/\sqrt{H}\quad P_u = P/H^{3/2}$$

Unit speed, unit discharge and unit power are used to compare turbines at one head and to predict performance at another head.

> Ex 1. Model turbine : $H = 6$ m, $N = 400$ rpm, $Q = 0.3$ m³/s, $P = 15$ kW. $N_u = 400/\sqrt{6} = 163.3$; $Q_u = 0.3/2.449 = 0.1225$; $P_u = 15/6^{1.5} = 1.02$. At $H = 36$ m : $N = 163.3 \times 6 =$ 980 rpm ; $Q = 0.1225 \times 6 =$ 0.735 m³/s ; $P = 1.02 \times 36^{1.5} =$ 220 kW.

Specific speed N_s

Speed of a geometrically similar turbine that develops 1 kW under a head of 1 m.

From $P \propto H^{5/2}/N^2$:

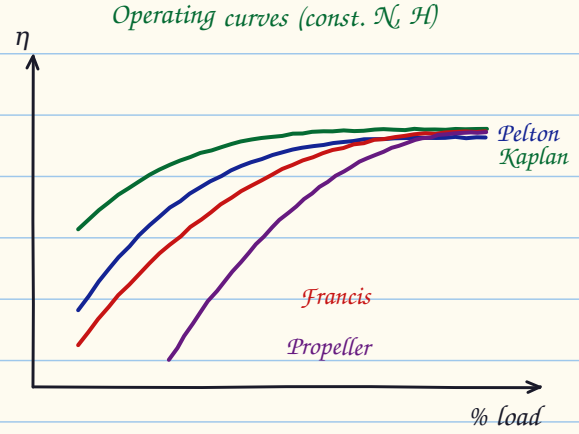
$$N_s = N\sqrt{P} / H^{5/4} \quad (N \text{ rpm}, P \text{ kW}, H \text{ m})$$

- It characterises the **type** of turbine independent of size : Pelton 10 – 60 (single jet) ; Francis 60 – 400 ; Kaplan/propeller 300 – 1000.
- Low N_s ⇒ high head, small discharge. High N_s ⇒ low head, large discharge, high speed.

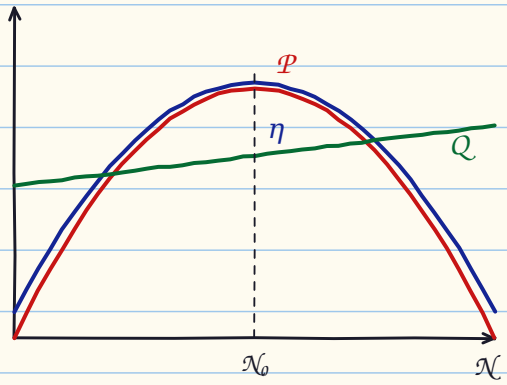
Ex 2. A turbine develops 10 000 kW at 200 rpm under 40 m head : $N_s = 200\sqrt{10000} / 40^{1.25} = 20000/100.6 = 199$ ⇒ Francis turbine.

Characteristic curves

Operating curves (const. N, H)



Main curves (constant head)



- **Main (constant head) curves** : N varied at constant H by changing the load/brake – plots of Q, P and η versus N ; efficiency peaks at the design (unit) speed N_0. For Pelton Q is practically constant (nozzle area fixed).
- **Operating (constant speed) curves** : N and H constant, gate opening varied – η vs % load. Kaplan & Pelton hold η over a very wide range ; Francis falls at part-load ; fixed-blade propeller falls steeply.
- **Constant-efficiency (Muschel / iso-efficiency) curves** : N_u vs Q_u (or H vs Q) contours of equal η ; the hill-diagram gives the full operating range and best-efficiency zone.

Governing of turbines

Maintaining a constant speed under varying load for the alternator frequency (50 Hz). A governor senses speed change and adjusts discharge through servo-motors (oil

pressure).

Turbine	Governing method	Remark
Pelton	Spear (needle) valve ; jet deflector / auxiliary nozzle	deflector avoids water hammer when load drops suddenly
Francis	Guide vane (wicket gate) opening	by-pass valve in extreme cases
Kaplan	Guide vanes + runner blade pitch	double regulation

Selection of the type of turbine

Head (m)	Type	Reason
> 250	Pelton (impulse)	high head, low flow, N_s low
60 – 250	Francis	medium head & flow
< 60	Kaplan / propeller	large discharge, high N_s
Fluctuating load	Pelton, Kaplan	flat efficiency curve
Very small heads (< 5 m)	Bulb / tubular turbines, water wheels	axial flow, run-of-river

Other factors : available discharge, cost, floor space, tail-water variation, cavitation limit, speed of generator, maintenance.

2. Cavitation in Turbines

When the local absolute pressure falls to the vapour pressure of water, vapour bubbles form (boiling at ordinary temperature) and are carried to a high-pressure region where they collapse suddenly, producing shock pressures (up to 1000 atm).

- **Where?** Reaction turbines : on the back (suction side) of runner blades near the outlet, draft tube inlet ; Pelton : at the nozzle tip and bucket splitter (rarely).
- **Effects** : pitting and erosion of the metal, noise & vibration, loss of power, drop in efficiency, reduced life.
- **Thoma's cavitation parameter** :

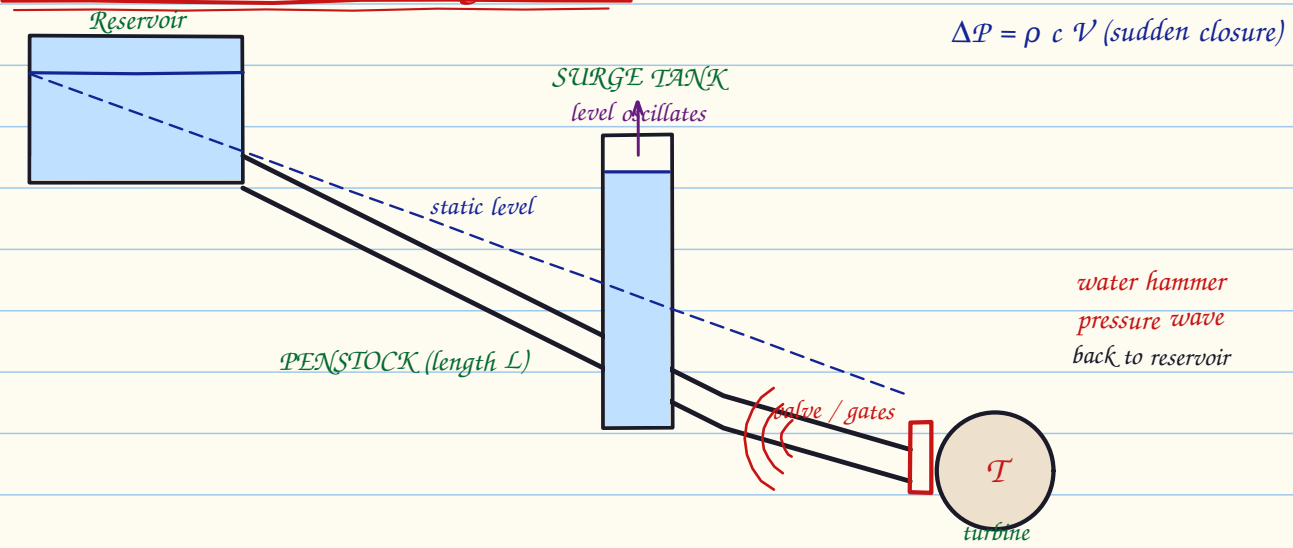
$$\sigma = (H_{atm} - H_v - H_s) / H = NPSH / H$$

- H_{atm} = atmospheric head (10.3 m water), H_v = vapour pressure head, H_s = suction (setting) height of runner outlet above tail water. The turbine is free from cavitation if $\sigma > \sigma_c$ (critical value from model tests, depends on N_s : $\sigma_c \approx 0.0432$

$(N_s/380.78)^2$ for Francis).

- Hence the maximum safe setting : $H_s \le H_{atm} - H_v - \sigma_c H$. For high-head Francis the runner must be set below tail water (negative H_s).
- **Prevention** : keep runner setting low, use cavitation-resistant materials (stainless steel, stellite), polished surfaces, correct draft-tube design, air admission, avoid part-load operation, good blade shapes.

3. Water Hammer & Surge Tank



Water hammer

When the flow in a long pipe is suddenly retarded (valve closed / gates shut on load rejection), the kinetic energy changes into pressure energy – pressure waves travel to and fro (speed $c \approx 1000$–1400 m/s) producing a severe pressure rise that may burst the pipe.

- Wave speed : $c = \sqrt{(K/\rho)} / \sqrt{(1 + K D/(E t))}$ (K bulk modulus, E Young's modulus of pipe, t thickness).
- **Sudden closure** (closing time $t_c \le 2L/c$) : rise in pressure $\Delta P = \rho c V$ (Joukowsky). **Gradual closure** ($t_c > 2L/c$) : $\Delta P \approx \rho L V / t_c$.
- Phases : (i) compression wave travels to reservoir (time L/c), (ii) reflected as release wave, (iii) negative pressure at the valve, (iv) returns ; period $4L/c$.

> Ex 3. Penstock $L = 1000$ m, $V = 2$ m/s, $c = 1200$ m/s : $\Delta P = 1000 \times 1200 \times 2 = 2.4$ MPa (≈ 245 m of water). Critical time $2L/c = 1.67$ s.

- **Prevention** : slow closing valves, surge tank, relief/pressure-regulating valves, air vessels, thick pipes, deflector in Pelton.

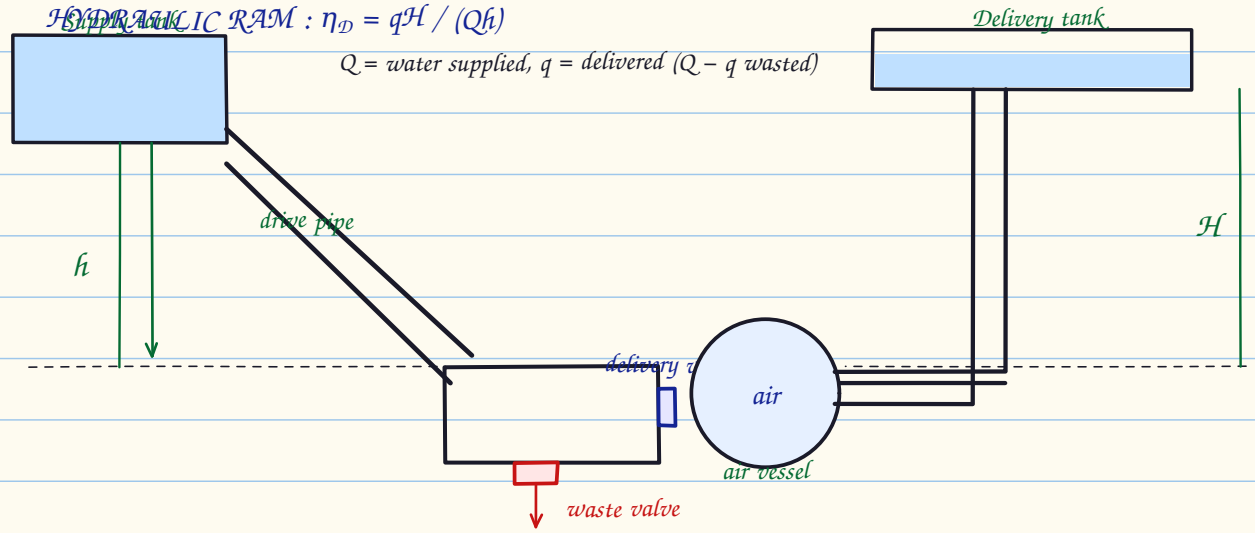
Surge tank

An open-top cylindrical storage tank connected to the penstock near the turbine.

- Functions : (a) during load rejection, absorbs the extra water (level rises) – prevents water hammer reaching the tunnel/reservoir ; (b) during load increase, supplies the extra water until the supply in the conduit is accelerated – so turbine isn't starved ; (c) reduces water-hammer pressure in the penstock section.
- Types : simple, restricted-orifice, differential (Johnson), closed (air-cushion), and with spillway. Water level in the surge tank oscillates (mass oscillation) and is damped by friction.

4. Hydraulic Systems

(a) Hydraulic ram



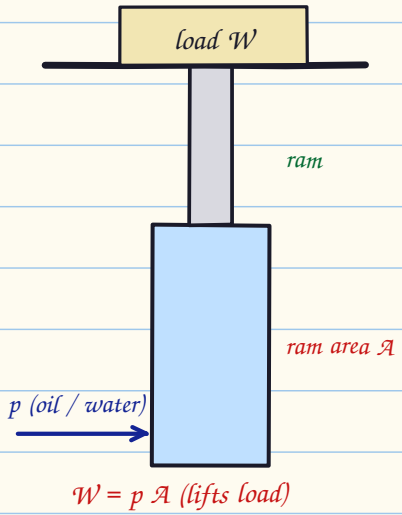
A device that lifts a small quantity of water to a great height using the energy of a large quantity falling through a small height – it works on the water hammer principle and needs no external power.

- Working : water flows from the supply tank down the drive pipe and out through the open waste (impulse) valve, accelerating. When velocity is large the waste valve snaps shut → sudden stoppage → pressure rise → opens the delivery valve and forces water into the air vessel and up the delivery pipe. Pressure falls, delivery valve closes, waste valve reopens and the cycle repeats (60 – 100 beats/min).
- Air vessel gives a steady delivery ; a snifter valve supplies the air it dissolves.

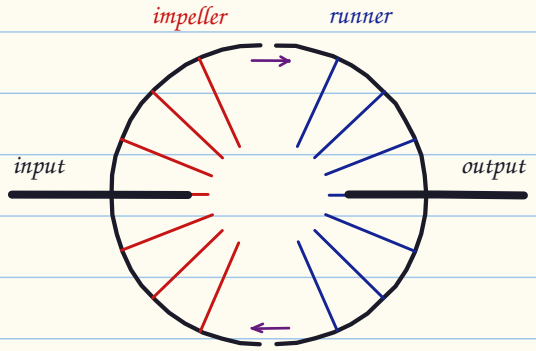
- Efficiency : D'Aubuisson $\eta_D = q\,H / (Q\,h)$; Rankine $\eta_R = q\,(H - h) / ((Q - q)\,h)$ where h = supply head, H = delivery head, Q = total flow from supply, q = delivered flow.

> Ex 4. $h = 2$ m, $H = 10$ m, $Q = 150$ l/min, $q = 10$ l/min : $\eta_D = (10 \times 10)/(150 \times 2) = 33$ % ; $\eta_R = 10(10 - 2)/(140 \times 2) = 28.6$ %.

(b) Hydraulic lift



FLUID COUPLING ($T_{in} = T_{out}$, slip 2–5 %)



- Used in buildings, garages, service stations. Liquid (oil) under pressure p acts on the ram of area A and lifts the cage : $W = p\,A$ (with pump pressure of 10–15 MPa a small ram lifts heavy loads — Pascal's law).
- Types : ***direct-acting*** (ram directly under the car ; travel = ram stroke), ***suspended*** (ram with pulley and chains – car travels farther than the ram, speed ratio 2:1 etc.), counter-weighted lifts for tall buildings. Pump, oil tank, check valves and control valves complete the system.

(c) Hydraulic coupling (fluid coupling)

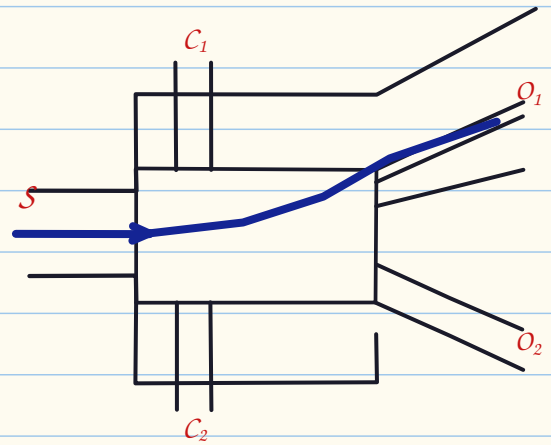
- Two identical halves – ***impeller*** (driving, pump) connected to the engine and ***runner*** (driven, turbine) connected to the load – each like a half toroid with radial vanes, filled with oil, with no mechanical contact. Oil thrown outward by the impeller enters the runner at the rim, transfers its angular momentum and returns inward.
- Torque input = torque output (no torque multiplication) ; slip $s = (N_1 - N_2)/N_1 \approx 2 - 5$ % ; efficiency $\eta = N_2/N_1 \approx 95 - 98$ %.
- Advantages : smooth start, cushions shocks, protects against overload, allows motor to reach full speed before load picks up. Used in buses, locomotives,

conveyors, ship propulsion.

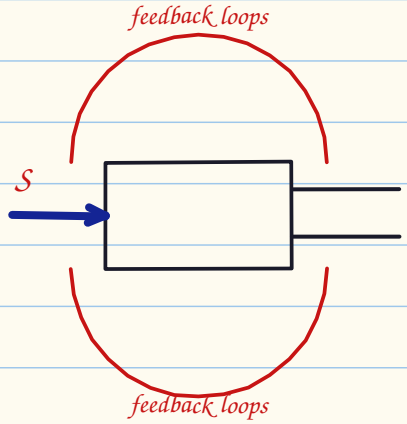
- **Torque converter** adds a fixed stator (reaction vanes) so that output torque exceeds input torque (multiplication up to 2–3×) – automatic transmissions. Other systems : hydraulic accumulator, intensifier, jacks, crane, press, brake.

5. Fluidics

(a) Bistable wall-attachment amplifier



(b) Fluidic oscillator



Fluidics = technology of using fluid flow (liquid/gas jets) for sensing, logic, amplification and control – devices with no moving parts.

- **Fluidic amplifiers** : a power jet from a supply nozzle S passes through an interaction chamber to two output channels. A small control flow at C_1 or C_2 deflects the jet to the other output. Types : bistable (wall-attachment, Coanda effect) – jet latches to a wall – flip-flop memory ; proportional – output varies with control flow ; turbulence amplifier – laminar jet triggered to turbulence ; vortex amplifier.
- **Fluidic sensors** : back-pressure sensor (proximity), interruptible-jet sensor (object breaks the jet), fluidic flow meter (vortex shedding), temperature (speed of sound / oscillation frequency), pressure sensors, level sensors.
- **Fluidic oscillators** : output jet fed back through feedback loops to the control ports, producing a continuous switching at frequency depending on loop length ($f \propto 1/\text{delay}$) – used as flow-meters, sprinklers, windshield washers, flow-modulation.

Advantages	Limitations
No moving parts – high reliability, long life, low maintenance	Low speed of response (few ms – kHz range limited by sound speed)

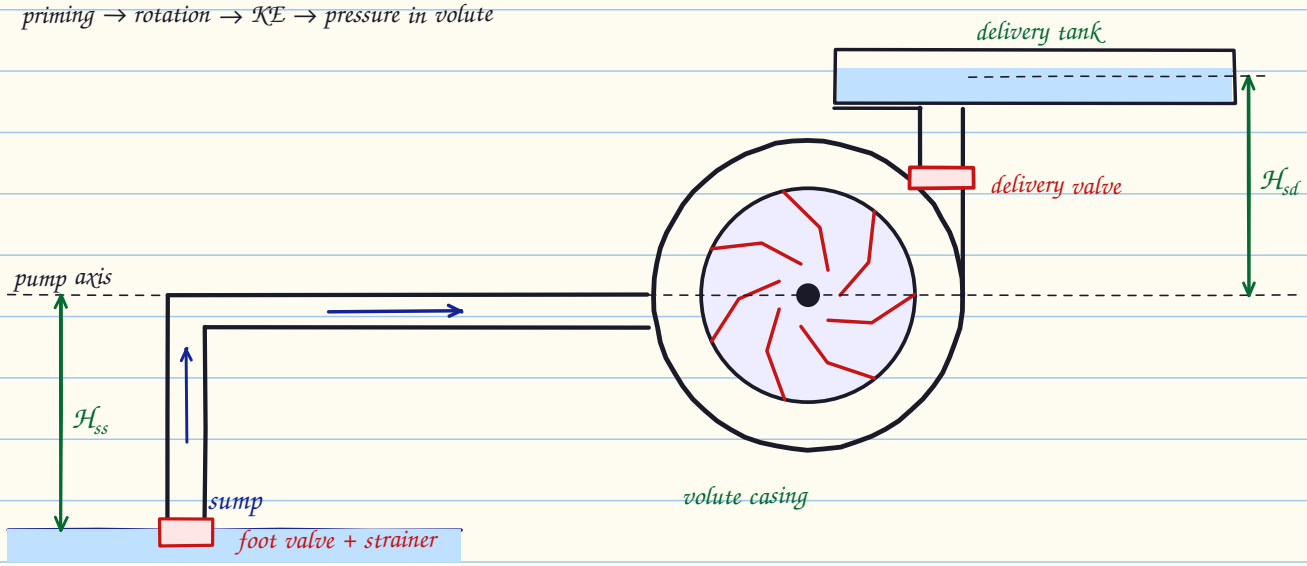
Advantages	Limitations
Work in hostile environments : high temp, radiation, explosive, vibration	Low gain & low signal levels ; need fluid supply and power continuously
Cheap, simple construction (moulded / etched plates)	Sensitive to dirt, viscosity & supply pressure changes ; noise
Fail-safe, inherently explosion-proof	Slower than electronics ; interfacing with other systems needed

Applications : logic and sequence control in factories, aircraft/missile control surfaces, fuel and engine control, flow measurement, medical (respirators, ventilators), automatic washers, vending, process industries, nuclear plants.

6. Centrifugal Pumps

CENTRIFUGAL PUMP

priming → rotation → KE → pressure in volute



A rotodynamic machine converting mechanical energy into hydraulic energy – the reverse of a radial-flow reaction turbine. Impeller rotating in a casing gives the water kinetic energy (centrifugal head) which is converted to pressure in the volute/diffuser.

Working & parts

- **Priming** : filling the suction pipe and casing with the liquid before starting – a centrifugal pump cannot lift air. As the impeller rotates, the liquid is thrown outward by centrifugal force, creating low pressure at the eye ; atmospheric pressure pushes more liquid up the suction pipe.
- Parts : impeller (backward-curved vanes), casing (volute, vortex, or diffuser type), suction pipe with foot valve and strainer, delivery pipe with delivery (regulating)

valve, shaft and bearings, stuffing box.

Classification

- By number of impellers : single-stage / multi-stage (for high head) ; by suction : single / double suction ; by impeller : open, semi-open, closed ; by casing : volute, vortex (with whirlpool chamber), diffuser (turbine pump).
- By head : low (< 15 m), medium (15 – 40 m), high (> 40 m) ; by specific speed : radial, mixed, axial ; by shaft : horizontal / vertical ; by priming : self-priming / ordinary.

Heads

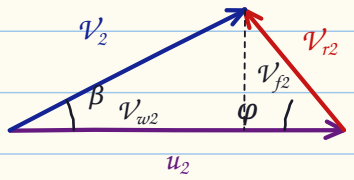
- Suction head H_s (centre-line above sump) ; delivery head H_d ; **static head** $H_{st} = H_s + H_d$.

$$H_m = H_{st} + h_{fs} + h_{fd} + V_d^2/2g \text{ (manometric head)}$$

- Manometric head = head imparted by impeller – losses inside pump = $(p_d - p_s)/\rho g + (V_d^2 - V_s^2)/2g + (z_d - z_s)$.

Work done by the impeller

OUTLET velocity triangle



Euler head per unit weight :

$H_e = (V_{w2}u_2 - V_{w1}u_1)/g$

radial entry ($\alpha = 90°$) : $V_{w1} = 0$

$V_{w2} = u_2 - V_{f2}\cot\varphi$

$\varphi < 90°$ backward curved vanes (usual)

$Q = \pi D_2 B_2 V_{f2}$

$$\text{Work/weight} = (V_{w2}u_2 - V_{w1}u_1)/g \text{ radial entry : } V_{w2}u_2/g$$

- Power given to water = $\rho Q(V_{w2}u_2 - V_{w1}u_1)$; $Q = \pi D_1 B_1 V_{f1} = \pi D_2 B_2 V_{f2}$.

Efficiencies

Efficiency	Definition	Typical
Manometric η_{man}	$H_m / (V_{w2}u_2/g) = gH_m/(V_{w2}u_2)$	65 – 90 %
Mechanical η_m	power at impeller / shaft power	90 – 95 %
Volumetric η_v	$Q/(Q + \text{leakage})$	95 – 98 %
Overall η_o	$\rho g Q H_m / SP = \eta_{man}\,\eta_m\,\eta_v$	60 – 85 %

- **Minimum starting speed** : delivery begins only when the centrifugal head ≥ manometric head : $(u_2^2 - u_1^2)/2g \geq \mathcal{H}_m \Rightarrow \mathcal{N}_{min} = 120\ \eta_{man}\ \mathcal{V}_{w2}\ \mathcal{D}_2 / [\pi\ (\mathcal{D}_2^2 - \mathcal{D}_1^2)]$.

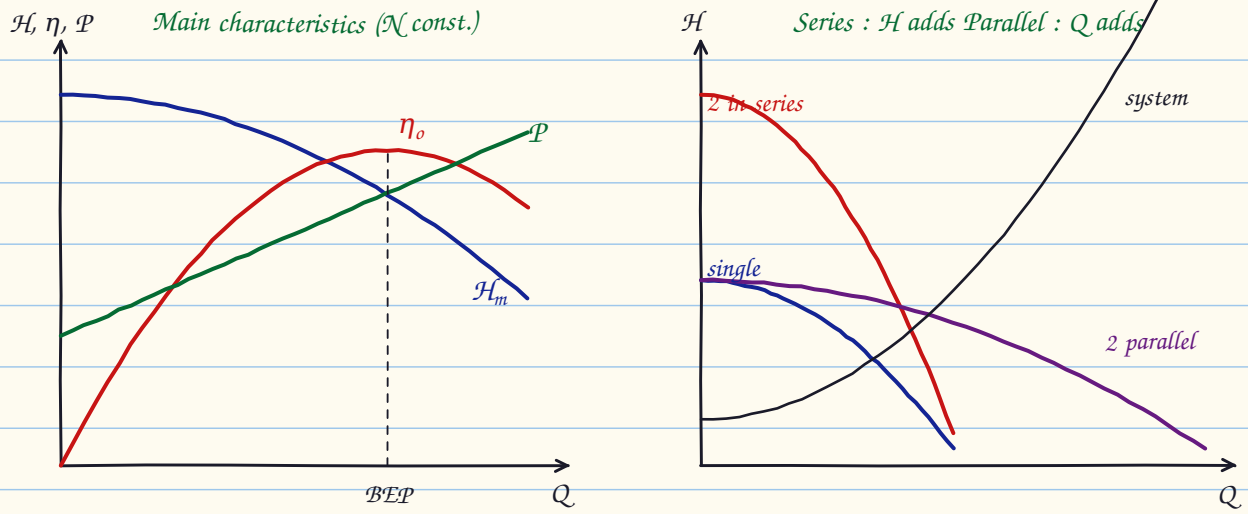
Ex 5. $\mathcal{D}_2 = 0.4$ m, $\mathcal{N} = 1450$ rpm, $\mathcal{V}_{f2} = 2.5$ m/s, $\varphi = 30°$, $\eta_{man} = 0.8$. $u_2 = \pi(0.4)(1450)/60 = 30.4$ m/s ; $\mathcal{V}_{w2} = u_2 - \mathcal{V}_{f2}/\tan\varphi = 30.4 - 4.33 = 26.0$ m/s ; $\mathcal{H}_m = \eta_{man}\ \mathcal{V}_{w2} u_2/g = 0.8 \times 26.0 \times 30.4/9.81 =$ 64.5 m.

Specific speed

$$\mathcal{N}_s = \mathcal{N}\sqrt{Q}\ /\ \mathcal{H}_m^{3/4} \quad (\mathcal{N}\ \text{rpm},\ Q\ m^3/s,\ \mathcal{H}_m\ m)$$

Speed of a geometrically similar pump that delivers 1 m^3/s against 1 m head. Radial (centrifugal) ≈ 10 – 80 ; mixed flow 80 – 160 ; axial (propeller) > 160. Similarity laws $\mathcal{H}/(\mathcal{N}^2\mathcal{D}^2)$, $Q/(\mathcal{N}\mathcal{D}^3)$, $\mathcal{P}/(\mathcal{N}^3\mathcal{D}^5)$ constant.

Pumps in series & parallel



- **Series** : delivery of first pump feeds the suction of second ⇒ same Q, heads add ($\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_2$) – used for very high heads (or multi-stage).
- **Parallel** : each pump draws from the sump and delivers to the same main ⇒ same $\mathcal{H}$, discharges add ($Q = Q_1 + Q_2$) – used for large discharge. With a steep system curve, two parallel pumps give less than twice the flow.

Performance characteristic curves

- **Main curves** (constant $\mathcal{N}$) : $\mathcal{H}_m$, $\mathcal{P}$ and η vs Q. $\mathcal{H}$ falls with Q, $\mathcal{P}$ rises, η rises to a maximum – the best efficiency point (BEP) – then drops. At $Q = 0$ (shut-off) power is least : start with delivery valve closed.

- **Operating curves** (constant head & efficiency ($\mathcal{N}$ varied)) ; **constant-efficiency curves** (Muschel) from tests at various speeds. **System curve** $\mathcal{H} = \mathcal{H}_{st} + K Q^2$; the intersection with the pump curve is the operating point.

Cavitation & NPSH

- Low suction pressure at the impeller eye (high lift, high speed, hot liquid) ⇒ cavitation ⇒ noise, vibration, fall of $\mathcal{H}$ & η, pitting.

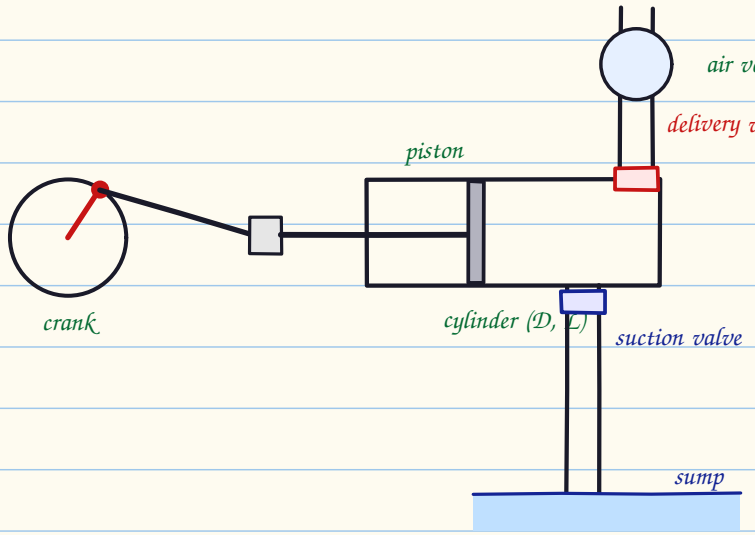
$$\mathcal{NPSH}_a = \mathcal{H}_{atm} - \mathcal{H}_v - \mathcal{H}_s - h_{fs} \text{ require } \mathcal{NPSH}_a > \mathcal{NPSH}_r$$

- $\mathcal{NPSH}_r$ is given by the manufacturer (test) ; Thoma's factor $\sigma = \mathcal{NPSH}_r/\mathcal{H}_m$. Remedies : reduce suction lift / friction, larger suction pipe, lower speed, double-suction impeller, inducers, cool the liquid.

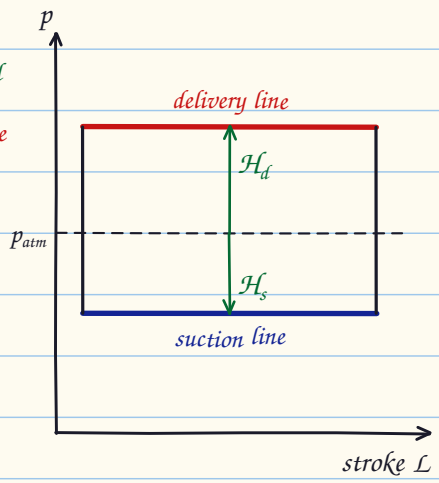
Ex 6. Sea-level pump : $\mathcal{H}_{atm}$ = 10.3 m, water at 30°C $\mathcal{H}_v$ = 0.43 m, suction lift 3 m, h_{fs} = 0.5 m : $\mathcal{NPSH}_a$ = 10.3 − 0.43 − 3 − 0.5 = 6.37 m.

7. Reciprocating Pumps

RECIPROCATING PUMP (single acting)



Indicator diagram (ideal)



Positive displacement pumps – the piston moving in a cylinder alternately draws liquid through the suction valve and forces it out through the delivery valve. Power to piston through a crank and connecting rod from a motor/engine.

Working & types

- Suction stroke : piston moves out, pressure in cylinder falls below atmospheric, suction valve opens, liquid enters ; delivery valve closed. Delivery stroke : piston moves in, suction valve closes, delivery valve opens.

- Types : **single-acting** (one delivery per revolution), **double-acting** (two), **duplex / triplex** (2 or 3 cylinders for smoother flow) ; **air vessels** on suction and delivery pipes.

Discharge, slip & power

$$Q_{th} = A\,L\,N / 60 \text{ (single)} \quad Q_{th} = 2\,A\,L\,N / 60 \text{ (double, neglecting rod)}$$

- Actual discharge $Q_{act} < Q_{th}$ owing to leakage, valve delay. **Slip** $= Q_{th} - Q_{act}$; % slip $= (Q_{th} - Q_{act})/Q_{th} \times 100$; coefficient of discharge $C_d = Q_{act}/Q_{th}$. *Negative slip* occurs if the delivery valve opens late / suction valve closes late (long suction pipe, high speed) – $Q_{act} > Q_{th}$.

$$\text{Work done/s} = \rho\,g\,Q\,(H_s + H_d) \quad \text{Power} = \rho\,g\,A\,L\,N\,(H_s + H_d) / 60 \text{ (single-acting)}$$

Ex 7. Single-acting : D = 150 mm, L = 300 mm, N = 60 rpm, H_s = 4 m, H_d = 20 m, Q_{act} = 5.0 l/s. $Q_{th} = 0.017671 \times 0.3 \times 60/60 = 5.30$ l/s ; slip = (5.30 – 5.00)/5.30 = 5.7 % ; power = 9810 × 0.0053 × 24 = 1.25 kW.

Indicator diagram

Plot of pressure in the cylinder (ordinate) against piston displacement (abscissa) – drawn by an indicator on the cylinder.

- **Ideal diagram** : a rectangle – suction line at $(H_{atm} - H_s)$ and delivery line at $(H_{atm} + H_d)$. Area = work done per cycle.
- **Effect of acceleration** : at the beginning of each stroke the piston accelerates, the liquid column must also be accelerated ⇒ extra pressure. *Acceleration head* $h_a = (L_p/g)(A/a)\,\omega^2\,r\cos\theta$ (maximum at beginning $= (L_p/g)(A/a)\omega^2 r$) ; at the end of stroke it is negative – pressure at the end of delivery stroke may fall to the vapour pressure ⇒ *separation* (cavitation) in the suction pipe.
- **Effect of friction** : friction head $h_f = (4\,f\,L_p/(2gD))\,(A\,v/a)^2$ varies as the square of the velocity, maximum in the middle of stroke ⇒ indicator diagram bulges. Actual diagram therefore has sloping lines and an area larger than the ideal.
- **Air vessels** : placed close to the cylinder on both suction and delivery sides ; they (i) smooth the flow, (ii) reduce acceleration head (by shortening the length of liquid that must be accelerated) and friction head, (iii) permit higher speed without separation, (iv) save power (≈ 39 % single-acting delivery side).

Reciprocating vs centrifugal pump

Item	Reciprocating	Centrifugal
Discharge	small, pulsating, fixed by speed	large, continuous, varies with head
Head	very high	low–medium (multi-stage for high)
Efficiency	high at all heads	max near BEP only
Priming	self-priming	needs priming
Size, cost, wear	large, heavy, many parts	compact, light, simple, cheap
Liquids	clean liquids ; viscous OK	clean, low-viscosity ; handles solids if open impeller

8. Quick Formula Sheet – Unit V

Item	Formula
Unit quantities	$N_u = N/\sqrt{H}$; $Q_u = Q/\sqrt{H}$; $P_u = P/H^{3/2}$
Specific speed (turbine)	$N_s = N\sqrt{P}/H^{5/4}$
Thoma factor	$\sigma = (H_{atm} - H_v - H_s)/H$
Water hammer	$\Delta P = \rho c V$ (sudden) ; $\rho L V/t$ (gradual)
Hydraulic ram	$\eta_D = qH/(Qh)$
Pump manometric head	$H_m = H_{st} + h_f + V_d^2/2g$
Pump efficiency	$\eta_o = \rho g Q H_m/SP$
Specific speed (pump)	$N_s = N\sqrt{Q}/H_m^{3/4}$
NPSH	$NPSH_a = H_{atm} - H_v - H_s - h_{fs}$
Reciprocating	$Q = ALN/60$; slip $= (Q_{th} - Q_{act})/Q_{th}$

9. Important Questions

- Define unit speed, unit discharge, unit power and specific speed of a turbine; derive N_s. Explain characteristic curves.
- Explain governing of Pelton, Francis & Kaplan turbines. How is a turbine selected for a given site?
- What is cavitation? Effects, Thoma's parameter, prevention in turbines and pumps.

- Explain water hammer & surge tank with sketches. Derive the pressure rise for sudden closure.
- Describe the hydraulic ram, hydraulic lift and fluid coupling with neat sketches; define efficiency of a ram.
- What is fluidics? Explain amplifier, sensor and oscillator; advantages, limitations & applications.
- Explain the working of a centrifugal pump; derive work done, manometric head, efficiencies, minimum starting speed, specific speed.
- Pumps in series/parallel; characteristic curves; NPSH. Reciprocating pump – discharge, slip, indicator diagram, air vessels.